

# WDW-7

## Profile-Clock Exhaustion and Constraint-Induced Cauchy-Germ Evolution

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Complete-record change and Hilbert-bundle companion response geometry

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**Exact scope.** The physical clock is the inherited function-valued WDW-4A gauge  $\phi(\cdot) = \tau$ , not its spatial mean. The accepted WDW-6 quotient is conditioned into the authenticated Cauchy-germ line, and the renormalized constraint induces the stated relational transport and complete-record change. The stronger full-Cauchy companion extension is retained separately; its nonzero Hamiltonian remains open.

## Abstract

This paper closes the source-generated WDW-7 clock-conditioning endpoint on the certified local Branch-A Wheeler–DeWitt target and separately constructs a stronger full-Cauchy companion extension. The construction begins by separating two categories that play different roles: the WDW-4A physical clock is the function-valued scalar gauge  $\phi(\cdot) = \tau \mathbf{1}$ , whereas the spatial mean of  $\phi$  is only a quotient-invariant analytic normal coordinate. No real-valued functional on configuration space can slice the full infinite-dimensional lapse distribution because its differential has a nonzero lapse kernel. The mean-zero complement remains in the residual germ, and the physical readout is evaluation at  $\chi = 0$ .

The accepted WDW-6 physical quotient is a rank-one Hilbert line with canonical coordinate  $I_6([u]) = u(F_{A,\mu}^{(0)})$ . The quotient-native analytic trace of the authenticated solution is the nonzero Cauchy-germ doublet  $Y_\tau^{(0)} = J_\tau F_{A,\mu}^{(0)}$ . Conditioning is the explicit isometry

$$\text{Cond}_\tau[u] = I_6([u])Y_\tau^{(0)}, \quad \mathcal{L}_\tau^C = \text{Ran Cond}_\tau,$$

with its positive form pushed forward from WDW-6. This construction retains the complete normal response and the transverse mean-zero scalar germ while adding no spectator directions or undeclared measure.

The accepted renormalized constraint is reapplied before any connection or Hamiltonian is defined. It gives

$$\hat{\mathcal{C}}^{\text{ren}} F_{A,\mu}^{(0)} = 0 \implies \dot{Y}_\tau^{(0)} = \mathbb{A}_\tau Y_\tau^{(0)} \implies \nabla_\tau^C = \partial_\tau - \mathbb{A}_\tau \quad \text{on } \mathcal{L}_\tau^C.$$

The induced two-parameter propagator is strongly continuous and unitary. Its covariant relational Hamiltonian is the bounded self-adjoint zero operator on the full generated fibre. For a differentiable section  $\xi_\tau = \alpha(\tau)Y_\tau^{(0)}$ , the relational Schrödinger equation holds exactly when  $\dot{\alpha} = 0$ , equivalently when  $\xi$  is propagated by the constraint-induced transport. The generated realization class is defined by antecedent clock, source, conditioning, no-spectator, form-compatibility, and complete-record requirements; the connection, propagator, Hamiltonian, and record values are then derived. The finite exhaustion eliminates external/lapse time, scalar-functional clocks, point-only traces, unproved frequency polarization, and undeclared spectator enrichment. Hence

$$|\text{Real}_7^{\text{gen}}/\sim_7| = 1, \quad \text{RayChange}_7 = \text{FALSE}.$$

The complete spatial-volume record is evaluated at the unique  $\phi(\cdot) = \tau \mathbf{1}$  intersection of the occupied classical constraint orbit. WDW-4A profile transversality and WDW-3 refoliation coherence give its descent; the spatial mean plays no role in that proof. Its derivative at the anchor is  $(3c_{\epsilon_*}/\epsilon_*) \text{Vol}(h_*) > 0$ . Therefore  $\text{RecordChange}_7 = \text{TRUE}$  and  $\text{Change}_7 = \text{TRUE}$ . The varying record is not asserted to be one fixed quantum operator intertwined across unequal clock slices.

The richer response construction retains the full analytic Cauchy-germ source. Frozen finite-cap increment coordinates yield an explicit compatible grade-zero form and actual weighted Hilbert adjoints. An unquotiented factorial history envelope realizes creation and prefix removal as genuine adjoints; the bound  $q(p) \leq n + 1$  gives convergent parity-resolved Dyson series; and a square-root-factorial Carleman estimate makes the symmetric generator  $i\hbar \mathcal{X}_0^{\text{pte}}$  essentially self-adjoint, equivalently making the underlying skew envelope essentially skew-adjoint. Companion propagation is unitary between transported Hilbert fibres. Unrestricted completed evaluation, a nonzero source-forced Hamiltonian on the enlarged full-Cauchy carrier, full operator-flat many-fingered propagation, and WDW-8 recovery remain open.

## Result at a glance

| Layer                           | Result                                                                                                                                                                                                        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Clock and authentication</b> | Unique inherited profile-clock class $\phi(\cdot) = \tau \mathbf{1}$ ; scalar-functional clock obstruction; invariant analytic normal splitting; unique minimal point character used only for authentication. |
| <b>Physical handoff</b>         | Canonical WDW-6 coordinate $I_6$ and an explicit isometric conditioning map into the authenticated Cauchy doublet.                                                                                            |

| Layer                                 | Result                                                                                                                                                                                   |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Generated conditional carrier</b>  | No-spectator line $\mathcal{L}_\tau^C = \mathbb{C}Y_\tau^{(0)}$ with the WDW-6 positive form pushed forward.                                                                             |
| <b>Constraint-induced evolution</b>   | State-independent companion factorization precedes the clock connection; strongly continuous unitary transport; bounded self-adjoint $H_{\text{rel}}^C = 0$ on the full generated fibre. |
| <b>Equation domain</b>                | $i\hbar\nabla_\tau^C\xi_\tau = 0$ holds precisely for propagated/parallel sections, not for every differentiable section.                                                                |
| <b>Physical classification</b>        | $ \text{Real}_7^{\text{gen}}/\sim_7  = 1$ , $\text{RayChange}_7 = \text{FALSE}$ , $\text{RecordChange}_7 = \text{TRUE}$ .                                                                |
| <b>Complete record</b>                | Spatial volume at the unique scalar-clock intersection of the occupied constraint orbit; no false cross-time fixed-operator intertwining.                                                |
| <b>Conditional datum</b>              | Full analytic germ restriction plus quotient-normal derivative, with the mean-zero scalar profile retained and physical evaluation at $\chi = 0$ ; both frequency sectors retained.      |
| <b>Residual operator</b>              | State-independent factorization of the complete renormalized WDW response.                                                                                                               |
| <b>Grade-zero form</b>                | Explicit compatible finite-cap form from frozen increment coordinates; exact radical and authenticated occupancy proved.                                                                 |
| <b>Source star representation</b>     | Registered daggers are weighted conjugate-transpose Hilbert adjoints on a dense common core, with maximal domains printed.                                                               |
| <b>Response envelope</b>              | Unquotiented factorial history Fock space; creation and annihilation are genuine adjoints.                                                                                               |
| <b>Evolution</b>                      | Explicit even/odd companion majorants and convergent response-history Dyson series.                                                                                                      |
| <b>Source evaluation</b>              | Exact algebraic and absolute-domain intertwining; unrestricted completed evaluation is open.                                                                                             |
| <b>Carleman closure</b>               | The genuine skew envelope is essentially skew-adjoint; the resulting Hamiltonian envelope is self-adjoint and contains the authenticated grade-zero vector in its domain.                |
| <b>Conditional geometry</b>           | Companion propagation is unitary between transported Hilbert fibres and defines a metric-compatible companion connection.                                                                |
| <b>Full-Cauchy Hamiltonian status</b> | No non-cancelling equality between the star envelope and the companion derivative is proved: <b>OPEN</b> .                                                                               |
| <b>Endpoint</b>                       | <b>PROFILE-CLOCK-CAUCHY-GERM-WDW7-CLOSED</b> ; the Hilbert-bundle companion theory is a stronger auxiliary extension.                                                                    |
| <b>Stronger endpoint</b>              | Mixed-profile operator flatness and WDW-8 recovery remain open.                                                                                                                          |

**Dependency order.** WDW-4A profile-clock transversality is fixed first. The scalar mean is then introduced only as analytic splitting skin, with its full mean-zero complement retained. The physical quotient supplies the canonical coordinate  $I_6$ ; quotient-native Cauchy-germ restriction forms  $Y_\tau^{(0)}$ ; and reapplication of the accepted renormalized constraint proves its companion equation. Only afterward are the generated line, pushed-forward form, connection, propagator, and covariant Hamiltonian constructed. Complete-record descent uses  $\phi(\cdot) = \tau\mathbf{1}$ , never  $T = \tau$ . The full-Cauchy extension begins separately; there the grade-zero form precedes every dagger, the genuine star representation precedes Carleman closure, and a nonzero full-Cauchy Hamiltonian is left open rather than obtained by cancellation.

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# Part I: Scope, contract, and theorem boundary

## 1. Original ARC obligation

The original WDW-7 contract asks how a state annihilated by the total Hamiltonian constraint can exhibit lawful change. On a certified physical clock patch it requests quotient-descended conditional states and observables, a compatible positive form, local conditional evolution, and record continuation, with no undeclared external parameter. Its strongest local form is

$$\hat{\mathcal{H}}\Psi = 0 \implies i\hbar \frac{\partial}{\partial T} \psi(q | T) = \hat{H}_{\text{rel}}(T) \psi(q | T), \quad (1)$$

on the physical domain and certified clock patch [1].

For a constraint second order in the clock, the non-lossy conditional datum is the Cauchy doublet

$$\psi(q | T) := Y_T(q) = \begin{pmatrix} \Psi|_T \\ \nabla_T \Psi|_T \end{pmatrix}. \quad (2)$$

A one-component positive-frequency reduction would require an additional polarization theorem and is not used.

Two scopes must be distinguished. The *source-generated physical endpoint* conditions the accepted rank-one WDW-6 quotient into the actual authenticated Cauchy doublet. Its carrier is the no-spectator line  $\mathbb{C}Y_\tau^{(0)}$ , but its section contains the full value/normal-response pair rather than a point-evaluation label. The renormalized constraint first supplies the companion equation; the induced clock connection, propagator, and covariant Hamiltonian are derived afterward. The *full-Cauchy extension* completes a larger response source capable of carrying data not generated by the accepted physical quotient. Its positive form, star envelope, and Hilbert-bundle propagation are proved, but a nonzero Hamiltonian identified with the full companion derivative remains open. The first scope attains the source-generated WDW-7 endpoint; the second is a stronger auxiliary theorem object.

## 2. One-way dependency order

The external ARC dependency remains

$$\text{WDW1} \rightarrow \text{WDW2} \rightarrow \text{WDW3} \rightarrow \text{WDW4A} \rightarrow \text{WDW5A} \rightarrow \text{WDW6A}^{\text{Pt}} \rightarrow \text{WDW7} \rightarrow \text{WDW8}.$$

WDW-1 through WDW-6 are frozen predecessor imports. WDW-8 and WDW-9 remain downstream. No WDW-7 construction may re-authorize a predecessor, replace the accepted renormalized operator, or import WKB data.

Within WDW-7 the proof order has four locked phases:

1. inherited function-valued clock quotient  $\rightarrow$  elimination of scalar-functional clock substitutes  $\rightarrow$  auxiliary analytic normal splitting  $\rightarrow$  point authentication and Cauchy-germ descent;
2. reapplication of the renormalized constraint  $\rightarrow$  state-independent companion factorization  $\rightarrow$  generated Cauchy conditioning  $\rightarrow$  pushed-forward form, connection, Hamiltonian, and propagator;
3. complete-record descent and relational change  $\rightarrow$  the source-generated realization quotient; and
4. compatible full-source form  $\rightarrow$  actual source adjoints  $\rightarrow$  unquotiented history star envelope  $\rightarrow$  graded evolution, Carleman closure, companion Hilbert-bundle transport, and the exact enlarged-carrier boundary.

### 3. Relation to the component results

The component studies record how the present theorem’s hypotheses and constructions were isolated [7, 8, 9, 10, 11]. They serve as provenance, not as substitutes for the proofs given here:

| Study                | Contribution retained                                                                                                                                                                                                           | Claim superseded or excluded                                                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WDW-7A</b>        | Certified clock patch, phase-only line continuation, and the nonconstant classical volume profile.                                                                                                                              | Identification of a point-evaluation section with constraint-derived conditional dynamics.                                                          |
| <b>WDW-7R</b>        | Quotient-native point authentication and the exact point-conditioning boundary.                                                                                                                                                 | Any claim that point data reconstruct second-order solution germs.                                                                                  |
| <b>WDW-7D</b>        | Full Cauchy doublet, observable typing, and state-independent companion factorization.                                                                                                                                          | Sampled feature metric, orbit-fitted generator, and frozen-address uniqueness.                                                                      |
| <b>WDW-7E</b>        | Source-response grading objective and transported-form question.                                                                                                                                                                | Unproved Picard, domain, unrestricted-form, and realization claims.                                                                                 |
| <b>WDW-7F</b>        | Companion path bound, parity majorants, and domain-separation objective.                                                                                                                                                        | Unconstructed grade-zero form, formal-dagger adjoint claim, resulting generator family, cancelling Hamiltonian equation, and <b>MANY-PHYSICAL</b> . |
| <b>Present paper</b> | WDW-6-compatible conditioning into the authenticated Cauchy line; constraint-induced transport; complete-record change; explicit source form, genuine star representation, and Carleman-closed full-Cauchy companion extension. | No nonzero full-Cauchy Hamiltonian is claimed.                                                                                                      |

### 4. Exact imported objects

Fix an accepted scale-cylinder record  $r_\mu$ . The imports are:

1. the compact, connected, boundaryless Branch-A metric–scalar chart and connected proper-redescription quotient [2, 5];
2. the WDW-3 first-class hypersurface-deformation closure and the WDW-4A occupied function-valued scalar clock gauge  $\phi(\cdot) = \tau \mathbf{1}$ , with  $\dot{\phi} = \epsilon_* > 0$  at the homogeneous anchor [3, 4];
3. the complete renormalized WDW-5/6 graph source  $\mathcal{V}_{A,\mu}^{\text{ren}}$ , its total operator endomorphisms, and the exact anomaly-free constraint AST [5, 6];
4. the complete joint kernel  $\mathcal{Z}_{A,\mu}$ , authenticated normalized member  $F_{A,\mu}^{(0)}(q_\star) = 1$ , positive physical quotient  $\mathcal{H}_6^{\text{pt}}$ , and canonical coordinate

$$I_6 : \mathcal{H}_6^{\text{pt}} \longrightarrow \mathbb{C}, \quad I_6([u]) = u(F_{A,\mu}^{(0)}); \quad (3)$$

5. finite caps, their fixed scalar encoding and echelon order, saturated task trees, stage-uniform extension, finite anticipation, and exact cap restriction identities from the sparse-jet and finite-anticipation results [12, 13]; and
6. first phase-jet sufficiency for regular classical histories and the warning that point profiles lose normal data [14, 15].

The predecessors do *not* supply a conditional Cauchy-germ line, a connection induced by the analytically normal-factorized constraint, a Hilbert form on the full Cauchy source, a Hilbert adjoint, a Fock completion, or a residual Hamiltonian. Each structure used below is therefore constructed at its declared scope.

## 5. Excluded inferences

The following implications are forbidden:

1. a spatial mean, fixed smearing, or other scalar functional of  $\phi \Rightarrow$  a slice of the full lapse/refoliation orbit;
2. an analytic normal coordinate  $T \Rightarrow$  an independently standing physical clock;
3. a normalized point character  $\Rightarrow$  a constraint-derived conditional evolution;
4. a varying scalar point profile  $\Rightarrow$  one fixed quantum operator intertwined between unequal clock slices;
5. every differentiable section of a zero-Hamiltonian line  $\Rightarrow$  a solution of the relational Schrödinger equation;
6. compatible locally convex cap seminorms  $\Rightarrow$  an unstated grade-zero Hilbert form;
7. preservation of an evaluation relation  $\Rightarrow$  preservation of Hilbert adjointness after factorial reweighting;
8. a formal dagger label  $\Rightarrow$  the maximal Hilbert adjoint;
9. left creation by a dagger letter  $\Rightarrow$  the adjoint of left creation;
10. a Carleman growth estimate  $\Rightarrow$  essential self-adjointness before symmetry is proved;
11. a transported metric on the enlarged Cauchy carrier  $\Rightarrow$  a nonzero source-forced Hamiltonian;
12.  $\Gamma = K - \mathbb{A}$  and  $H = i\hbar K \Rightarrow$  a Hamiltonian derivation; and
13. a continuum of auxiliary positive forms  $\Rightarrow$  a continuum of physical Hamiltonian realizations.

## 6. What is proved here

The proof supplies:

1. the exact WDW-4A function-valued profile clock and an exhaustion theorem eliminating external, lapse, foliation, and scalar-functional substitutes in the frozen handoff class;
2. a quotient-invariant analytic mean/mean-zero splitting in which the mean is coordinate skin and the entire mean-zero scalar germ is retained;
3. the quotient-native Cauchy trace of the authenticated WDW-6 solution;
4. an explicit conditioning map from the accepted WDW-6 physical quotient into the generated Cauchy line, with the physical form pushed forward isometrically;
5. the actual analytically normal-factorized companion equation derived from the renormalized constraint before any connection or Hamiltonian is defined;
6. the induced Hermitian connection, strongly continuous unitary propagator, full-domain bounded self-adjoint zero covariant Hamiltonian, and the exact propagated-section equation;
7. a complete relational volume record on the occupied classical constraint orbit, including proper-redescription, lapse-reparametrization, and refoliation descent;
8. a noncircular source-generated realization quotient and its singleton cardinality;
9. an explicit compatible grade-zero positive form with exact radical for the stronger full-Cauchy extension;
10. a dense finite-cap core and the weighted matrix formula for every actual Hilbert adjoint;

11. an unquotiented response-history Fock envelope in which creation and annihilation are actual adjoints;
12. exact algebraic evaluation and an explicit absolute source-majorized domain;
13. the retained explicit even/odd evolution majorants;
14. a reference Carleman theorem for the genuine skew star envelope;
15. authenticated-state occupancy of the closed star-envelope domain;
16. transported Hilbert fibres and the companion connection  $\partial_\tau - \mathbb{A}_\tau$ ; and
17. a proof that these enlarged-carrier data do not identify the star envelope with the companion derivative.

**Audit finding 6.1** (Why support sources are not substituted for the missing theorem). The boundary-trace source fixes role anchors, not positive forms [18]. The transport source stops at comparison geometry [19]. The curvature source makes flatness an additional burden [20]. Fixed-base discipline forbids hidden form or domain import [21]. Self-adjoint extension selection applies only after a positive form and symmetric operator have been constructed [22]. None of these sources supplies the Cauchy conditioning map, companion restriction, complete-record descent, or full-source Hilbert construction proved here.

**Scope lock.** The physical clock is exactly the inherited WDW-4A profile gauge; the spatial mean is only an analytic normal coordinate and may not acquire clock standing. The primary theorem concerns the source-generated Cauchy-germ line obtained from the accepted WDW-6 quotient and authenticated solution. It does not claim uniqueness among clocks from different sources or branches, nor among every enriched conditional carrier. The larger full-Cauchy Hilbert-bundle construction is stated separately, and its nonzero-Hamiltonian gate remains open.

## Part II: Physical profile clock and analytic normal splitting

### 7. The inherited physical clock

On the WDW-4A occupied datum,

$$z_{\epsilon_*} = (h_*, \pi_*; \phi_* = 0, p_{\phi,*} = \epsilon_* \sqrt{h_*}), \quad \epsilon_* > 0,$$

with

$$K_{ij,*} = c_{\epsilon_*} h_{ij,*}, \quad c_{\epsilon_*} = \sqrt{1 + \frac{\kappa \epsilon_*^2}{6}} > 0,$$

the accepted normal convention gives  $\dot{\phi} = \epsilon_*$ . The scalar gradient is timelike at the anchor and therefore on a common local spacetime neighborhood. Its future-directed normalized flow

$$V_\phi = \frac{\nabla \phi}{g^{-1}(d\phi, d\phi)}, \quad V_\phi(\phi) = 1,$$

licenses the WDW-4A level-set readout  $\phi(\cdot) = \tau$ ; it does not prove global monotonicity.

**Definition 7.1** (WDW-7 physical clock-use package). The inherited physical clock object is

$$\mathcal{D}_{\text{clk}}^{(7)} = (C_\phi, U_\phi, A_\phi, Q_\phi, R_\phi, K_\phi, \mathcal{F}_{\text{clk}}; \mathcal{Z}_{A,\mu}, F_{A,\mu}^{(0)}), \quad (4)$$

where  $C_\phi$  is the spacetime scalar profile;  $U_\phi$  is local ordering by its spacelike level sets;  $A_\phi$  fixes the source history, anchor, and orientation;  $Q_\phi$  is the connected proper-redescription and guarded characteristic quotient;  $R_\phi$  contains the scalar and relational readouts at  $\phi(\cdot) = \tau$ ;  $K_\phi$  is continuation along the certified scalar flow; and  $\mathcal{F}_{\text{clk}}$  is the first-failure boundary. The final coordinates bind the use to the accepted WDW-6 joint source and authenticated member.

The clock has standing only as this use-relative quotient package [17]. The same lineage terminates at the first occurrence of null or spacelike scalar gradient,  $d\phi = 0$ , vanishing oriented normal speed, loss of characteristic transversality, multiple guarded intersections, readout or record failure, generalized-KID isotropy, source/attachment/branch change, or termination of the licensed local development. A later valid patch starts a new lineage; it does not restore standing to the failed prefix.

## 8. Why no scalar functional is the physical refoliation clock

For a lapse-shift descriptor  $(N, X)$ , the accepted canonical scalar response is

$$\delta_{(N,X)}\phi = \mathcal{L}_X\phi + N \frac{p_\phi}{\sqrt{h}}. \quad (5)$$

At the homogeneous anchor,  $D\phi_* = 0$  and  $p_{\phi,*}/\sqrt{h_*} = \epsilon_*$ , so

$$\delta_{(N,0)}\phi = \epsilon_* N. \quad (6)$$

**Lemma 8.1** (Scalar-functional clock obstruction). *Let  $S$  be any continuously differentiable real-valued functional on the regular canonical chart. Its infinitesimal response along normal characteristic directions is the continuous linear functional*

$$\Lambda_S : C^\infty(\Sigma) \longrightarrow \mathbb{R}, \quad \Lambda_S(N) = dS_{z_{\epsilon_*}}(X_{(N,0)}).$$

*Because the lapse space is infinite dimensional,  $\Lambda_S$  has a nonzero kernel. Hence the single equation  $S = \tau$  cannot be a local slice of the full normal characteristic leaf at the anchor.*

*Proof.* A linear map from an infinite-dimensional vector space to a one-dimensional vector space is not injective. Choose  $N \neq 0$  in  $\ker \Lambda_S$ . The WDW-4A no-KID theorem and eq. (6) show that  $X_{(N,0)}$  is a genuine normal characteristic direction, not a spatial-redescription zero. It is tangent to  $S = \tau$ , so that level condition does not fix the full refoliation freedom and cannot supply the unique guarded intersection required by the inherited clock contract.  $\square$

This obstruction includes the spatial mean of  $\phi$ , any fixed smearing of  $\phi$ , and any other scalar field-space coordinate. It does not obstruct the function-valued gauge: eq. (6) is an isomorphism from lapse profiles to scalar profiles at the anchor, and the accepted WDW-4A implicit-function/flow-box theorem therefore gives the local slice

$$\phi(\cdot) = \tau \mathbf{1} \quad (7)$$

after the WDW-2 proper spatial quotient.

**Theorem 8.2** (W7.1: physical clock exhaustion). *Within the frozen WDW-4A handoff, the local physical-clock comparison class is nonempty and has one class up to the admitted orientation-preserving clock reparametrizations:*

$$\left| \text{Clk}_7^{\text{phys}} / \sim_{\text{clk}} \right| = 1. \quad (8)$$

*Its representative is the profile gauge eq. (7). External time and lapse/foliation labels fail the inherited quotient-use gate; every scalar-valued functional fails lemma 8.1; a monotone relabeling of  $\phi$  is an allowed presentation change; and a clock with a different source, readout, orientation, or failure profile is a different lineage rather than a same-scope rival.*

*Proof.* WDW-3 classifies bulk lapse and foliation as presentation data and WDW-4A constructs the populated profile-clock package, proving nonemptiness. Lemma 8.1 excludes the entire scalar-functional family at one typed gate. The WDW-4A overlap theorem identifies exactly the orientation-preserving reparametrizations that preserve the clock-use tuple. Fixed-domain closure then prevents an excluded clock from acquiring standing through a later analytic coordinate, conditioning map, or Hamiltonian construction [25, 26]. These cases exhaust the frozen handoff class; the result is not a claim that every conceivable internal clock in a different source or branch is equivalent.  $\square$

## 9. An auxiliary invariant normal coordinate

For analytic factorization of the functional differential operator, define on the positive-metric configuration chart

$$T(h, \phi) = \frac{1}{\text{Vol}(h)} \int_{\Sigma} \phi d\mu_h. \quad (9)$$

The notation  $T$  denotes an auxiliary quotient coordinate, not the physical clock. It is smooth in the accepted Sobolev range and invariant under connected proper spatial redescriptions. Let

$$\pi_{\text{pr}} : \mathcal{U}_{\text{reg}} \longrightarrow \mathcal{Q}_{\text{reg}}$$

be the WDW-2 regular local quotient submersion. Local descent gives a unique smooth function  $\bar{T}$  with  $T = \bar{T} \circ \pi_{\text{pr}}$ . Along the homogeneous anchor history,  $T = \phi$  and  $d\bar{T}([\dot{q}_*]) = \epsilon_* \neq 0$ .

**Lemma 9.1** (Invariant analytic splitting). *There is a source-preserving neighborhood  $\bar{\mathcal{U}}_7$  of  $[q_*]$  in the completed regular local quotient on which  $d\bar{T}$  is surjective and a smooth flow-box diffeomorphism*

$$\bar{\kappa} : \bar{\mathcal{U}}_7 \longrightarrow I_7^{\text{an}} \times \mathcal{Y}, \quad \bar{T} \circ \bar{\kappa}^{-1}(\tau, y) = \tau, \quad (10)$$

with  $0 \in I_7^{\text{an}}$ .

*Proof.* The target of  $\bar{T}$  is one dimensional. Since  $d\bar{T}_{[q_*]}([\dot{q}_*]) = \epsilon_* \neq 0$ , its differential is split-surjective. WDW-2 supplies the completed Hilbert local quotient chart. The Banach submersion theorem at a sufficiently high Sobolev stage gives the product chart, and the accepted compatible-stage restriction gives the inverse-limit chart. No section of  $\pi_{\text{pr}}$  is selected.  $\square$

Write locally

$$\phi = T\mathbf{1} + \chi, \quad \int_{\Sigma} \chi d\mu_h = 0. \quad (11)$$

Let  $\iota_{\tau}(y) = \bar{\kappa}^{-1}(\tau, y)$  and  $\partial_T = \bar{\kappa}_*^{-1}(\partial_{\tau})$ . The residual coordinate  $y$  retains the complete mean-zero profile  $\chi$  and every non-clock quotient direction. The analytic hypersurface  $T = \tau$  is therefore not identified with the physical gauge. The latter is the submanifold

$$\mathcal{S}_{\tau}^{\text{phys}} = \{\iota_{\tau}(y) : \chi(y) = 0\}, \quad (12)$$

corresponding to  $\phi(\cdot) = \tau\mathbf{1}$ . The Cauchy source below retains the germ in  $\chi$  before evaluation at  $\chi = 0$ ; consequently all transverse scalar-response terms remain available to the renormalized constraint.

## 10. Common certified interval and no-backflow

Let  $I_7$  be the anchor component of the intersection of the inherited WDW-4A profile-clock interval and  $I_7^{\text{an}}$ , further restricted by retention of the same source, branch, quotient and renormalized-operator record, nonvanishing of the clock-normal principal coefficient, and continuity of every finite-cap coefficient required below.

The interval  $I_7$  is nonempty and open. The physical level sets are those of eq. (7);  $T$  and  $\partial_T$  provide only the analytic normal/tangential decomposition used by the operator. By the AASC role classification,

changing the analytic complement or quotient chart while covariantly transporting the Cauchy germ is skin. Promoting that complement, the lapse, or the mean functional to a new physical clock would introduce an unlicensed same-scope selector and is forbidden by theorem 8.2.

**Proposition 10.1** (No external-time or analytic-coordinate backflow). *Neither  $\tau$ ,  $T$ , nor a lapse profile is an external evolution parameter. The parameter  $\tau$  is the readout value of the admitted scalar profile on eq. (7);  $T$  is an auxiliary normal coordinate agreeing with that readout only on  $\chi = 0$ ; and lapse remains a deformation multiplier. None may retro-authorize the clock-use package from a downstream propagator or Hamiltonian.*

## Part III: Cauchy conditioning and constraint-induced line evolution

### 11. Point authentication without a dynamical category switch

Let  $q(\tau)$  be any smooth raw presentation of the occupied classical quotient history and let

$$\delta_{q(\tau)}(F) = F(q(\tau)), \quad F \in \mathcal{V}_{A,\mu}^{\text{ren}}.$$

The compact-open Bastiani seminorms contained in the WDW-5/6 graph topology make  $\tau \mapsto \delta_{q(\tau)}$  smooth in the strong continuous dual. Let  $Q_6$  denote the accepted WDW-6 physical-radical quotient and put

$$e_\tau = Q_6 \delta_{q(\tau)}, \quad c(\tau) = F_{A,\mu}^{(0)}(q(\tau)). \quad (13)$$

**Proposition 11.1** (Quotient-native point authentication). *For every connected proper presentation  $\tilde{q}(\tau) = g_\tau q(\tau)$ ,*

$$Q_6 \delta_{\tilde{q}(\tau)} = Q_6 \delta_{q(\tau)}.$$

*The class  $e_\tau$  is smooth, scale compatible, and nonzero near the anchor. Where  $c(\tau) \neq 0$ , the normalized class  $c(\tau)^{-1}e_\tau$  is mapped to 1 by the canonical WDW-6 coordinate  $I_6$ .*

*Proof.* The complete momentum constraint annihilates  $F_{A,\mu}^{(0)}$ . Integrating its infinitesimal action along a connected proper isotopy gives  $F_{A,\mu}^{(0)}(gq) = F_{A,\mu}^{(0)}(q)$ . Since the WDW-6 physical form is

$$B_{A,\mu}(u, v) = \overline{u(F_{A,\mu}^{(0)})}v(F_{A,\mu}^{(0)}),$$

the difference  $\delta_{q(\tau)} - \delta_{\tilde{q}(\tau)}$  lies in its radical. This proves representative independence after, but not before, the exact physical quotient. Strong-dual smoothness follows from the uniform compact-open derivative estimate along the smooth clock path. Finally  $c(0) = F_{A,\mu}^{(0)}(q_\star) = 1$  and  $I_6(c(\tau)^{-1}e_\tau) = 1$ .  $\square$

Let  $\mathcal{A}_{x_\tau}^{\text{germ}}$  be the local star-algebra of quotient-invariant smooth germs at the clock-selected quotient point  $x_\tau = [q(\tau)]_{\text{pr}}$ .

**Lemma 11.2** (Minimal point-character exhaustion). *Every continuous unital complex-linear multiplicative star-character  $\chi_\tau : \mathcal{A}_{x_\tau}^{\text{germ}} \rightarrow \mathbb{C}$  is evaluation:*

$$\chi_\tau([O]_{x_\tau}) = O(x_\tau). \quad (14)$$

*Thus the minimal point-authentication class is unique.*

*Proof.* The germ algebra is local. Its unique maximal ideal consists of germs vanishing at  $x_\tau$ , because every germ with nonzero value has a smooth reciprocal on a sufficiently small neighborhood. The kernel of a unital character is maximal, hence it is this vanishing ideal, which forces eq. (14).  $\square$

The point character authenticates the clock-selected source class; it is not used as the conditional state carrier. In particular, no varying family  $O(x_\tau)I$  is asserted to be one fixed physical operator intertwined between different clock slices.

## 12. Analytic Cauchy-germ restriction and physical evaluation

Let

$$\mathcal{V}_{\text{pr}}^{\text{ren}} = \{F \in \mathcal{V}_{A,\mu}^{\text{ren}} : \widehat{\mathcal{D}}^{\text{ren}}[X]F = 0 \text{ for every connected proper generator } X\}.$$

WDW-2 local observable descent and WDW-6 constraint annihilation give each  $F \in \mathcal{V}_{\text{pr}}^{\text{ren}}$  a unique descended function  $\bar{F}$  on  $\bar{\mathcal{U}}_7$ . Let  $\mathcal{W}_\tau$  be the residual smooth graph source of germs near  $\chi = 0$  obtained by pulling this descended WDW-6 grammar to  $\mathcal{Y}$ . Define

$$R_\tau F = \bar{F} \circ \iota_\tau, \quad P_\tau F = (\partial_T \bar{F}) \circ \iota_\tau, \quad J_\tau F = (R_\tau F, P_\tau F). \quad (15)$$

The physical profile-slice evaluation is the separate map

$$\text{Ev}_\tau^\phi(R_\tau F) = (R_\tau F)|_{\chi=0}, \quad \phi(\cdot) = \tau \mathbf{1}. \quad (16)$$

Thus  $J_\tau F$  retains the complete mean-zero scalar germ required by the operator, while the physical clock readout uses only the WDW-4A submanifold eq. (12). The field-space coordinate  $T$  supplies analytic normal splitting but carries no independent clock standing. This Cauchy trace is unrelated to the bivalent role-anchor object called “boundary trace” in [18].

**Theorem 12.1** (W7.2: strong graph smoothness of Cauchy-germ restriction). *For every  $F \in \mathcal{V}_{\text{pr}}^{\text{ren}}$ ,  $R_\tau F$  and  $P_\tau F$  belong to the residual germ graph source. The maps  $R_\tau$ ,  $P_\tau$ ,  $J_\tau$ , and  $\text{Ev}_\tau^\phi$  are continuous on their declared germ domains, and  $\tau \mapsto J_\tau F$  is smooth in every admitted finite-cap graph seminorm. Moreover,*

$$\partial_\tau R_\tau F = P_\tau F. \quad (17)$$

*Proof.* WDW-2 identifies invariant smooth functions with smooth functions on the regular local quotient; WDW-6 graph covariance makes that identification continuous on the proper-constraint kernel. Composition of  $\bar{F}$  with the smooth quotient embedding is continuous for compact-open Bastiani seminorms. The successor graph topology additionally records every finite word in the complete renormalized operators. Quotient-coordinate pullback turns each such word into a finite sum of residual differential words with smooth coefficients, so each pulled-back graph seminorm is bounded by finitely many inherited graph seminorms on the compact image of a cap. The chain rule gives eq. (17); iteration gives the finite Faà di Bruno expressions for all clock derivatives. The finite-anticipation theorem makes the bound stage-uniform at every requested finite jet degree.  $\square$

## 13. Proper-redescription descent

Let  $g$  be a connected proper arrow between two raw presentations. The complete momentum constraint annihilates  $F \in \mathcal{Z}_{A,\mu}$ , so integration along a proper isotopy gives  $F(gq) = F(q)$  and produces the unique  $\bar{F}$ . Normal differentiation is taken downstairs by  $\partial_T$  in the quotient flow box and therefore has no representative dependence to remove.

**Theorem 13.1** (W7.3: quotient-native Cauchy-germ conditioning). *For  $F \in \mathcal{Z}_{A,\mu}$ ,  $J_\tau F$  is intrinsic to the regular local proper quotient and therefore defines the quotient-native Cauchy germ*

$$[J_\tau F]_{\text{pr}} \in \mathcal{C}_\tau := J_\tau(\mathcal{Z}_{A,\mu}) \subset \mathcal{W}_\tau \oplus \mathcal{W}_\tau. \quad (18)$$

*Any two raw-presentation pullbacks give this same datum, its physical evaluation eq. (16) lies on the inherited profile-clock slice, and both are compatible with WDW-6 scale identity transport.*

*Proof.* Both raw presentations pull back the same unique descended function  $\bar{F}$  and the same quotient-normal derivative. This proves representative independence without choosing a global or local section of  $\pi_{\text{pr}}$ . Restriction to  $\chi = 0$  then implements the already certified physical profile gauge; it does not turn the analytic mean coordinate into a clock. The scale-cylinder theorem fixes the source, operator family, normalized solution, and graph topology literally, so it fixes every coordinate used in eq. (15).  $\square$

Large spatial transformations, sector changes, and profile changes outside the connected proper group remain unquotiented.

## 14. Constraint-induced companion factorization

On the proper-constraint kernel, decompose the lapse grammar into the constant mode  $N_0 = \mathbf{1}$  fixed by the analytic splitting and its residual profile complement. The  $N_0$  component of the complete renormalized constraint has the quotient normal form

$$\hat{\mathcal{C}}_{N_0}^{\text{ren}} = -\hbar^2(a_\tau \partial_T^2 + B_\tau \partial_T) + \Theta_\tau. \quad (19)$$

Put

$$C_\tau = \hbar^{-2} a_\tau^{-1} \Theta_\tau, \quad D_\tau = -a_\tau^{-1} B_\tau, \quad \mathbb{A}_\tau = \begin{pmatrix} 0 & I \\ C_\tau & D_\tau \end{pmatrix}. \quad (20)$$

Here  $C_\tau$  has response cost at most two,  $D_\tau$  has response cost at most one, and the upper-right identity has cost zero. These costs are read from the serialized WDW-6 AST; no density, ordering, connection, regulator, counterterm, retained potential, mean-zero scalar response, or residual lapse constraint is deleted. The choice  $N_0 = \mathbf{1}$  fixes the normal coordinate of the analytic decomposition; it is not promoted to the physical clock or a preferred foliation.

**Theorem 14.1** (W7.4: state-independent residual factorization). *For every  $F \in \mathcal{Z}_{A,\mu}$ ,*

$$\frac{d}{d\tau} J_\tau F = \mathbb{A}_\tau J_\tau F, \quad \text{Ker } J_\tau \cap \mathcal{Z}_{A,\mu} \subseteq \text{Ker} \left( \frac{d}{d\tau} J_\tau \right). \quad (21)$$

*The operator and its source graph domain do not depend on the authenticated member  $F_{A,\mu}^{(0)}$ .*

*Proof.* The first component is the slice chain rule. Restricting  $\hat{\mathcal{C}}_{N_0}^{\text{ren}} F = 0$  and solving for the second normal derivative gives the lower component. If  $J_\tau F = 0$ , both the residual function and its first normal derivative vanish, so every tangential derivative in the lower block vanishes. The constant-normalization obstruction is retained:  $\Theta_\tau \mathbf{1}$  still carries the potential term [16].  $\square$

## 15. Generated Cauchy-line conditioning and evolution

The accepted solution defines

$$Y_\tau^{(0)} = J_\tau F_{A,\mu}^{(0)}. \quad (22)$$

The accepted normalization gives  $(R_0 F_{A,\mu}^{(0)})([q_\star]) = 1$ , so  $Y_0^{(0)} \neq 0$ . Define the maximal generated-line pass interval

$$I_7^C = \text{Comp}_0\{\tau \in I_7 : Y_\tau^{(0)} \neq 0\}. \quad (23)$$

For  $\tau \in I_7^C$ , set

$$\text{Cond}_\tau : \mathcal{H}_6^{\text{pt}} \longrightarrow \mathcal{C}_\tau, \quad \text{Cond}_\tau[u] = I_6([u])Y_\tau^{(0)}, \quad (24)$$

$$\mathcal{L}_\tau^C = \text{Ran Cond}_\tau = \mathbb{C}Y_\tau^{(0)}, \quad (25)$$

$$\langle \text{Cond}_\tau u, \text{Cond}_\tau v \rangle_\tau^C = \langle [u], [v] \rangle_6. \quad (26)$$

Because  $I_6$  is an isometric isomorphism, eq. (26) is well defined and makes  $Y_\tau^{(0)} = \text{Cond}_\tau(I_6^{-1}1)$  a unit vector. No ambient full-source norm is used.

Define

$$U^C(\tau, \sigma)(\alpha Y_\sigma^{(0)}) = \alpha Y_\tau^{(0)}, \quad (27)$$

$$\nabla_\tau^C \xi_\tau = (\partial_\tau - \mathbb{A}_\tau) \xi_\tau, \quad (28)$$

$$H_{\text{rel}}^C(\tau) = 0, \quad \text{Dom } H_{\text{rel}}^C(\tau) = \mathcal{L}_\tau^C. \quad (29)$$

**Theorem 15.1** (W7.5: WDW-6-compatible generated Cauchy-line evolution). *The interval  $I_7^C$  is nonempty and open. The maps  $\text{Cond}_\tau$  are unitary isomorphisms from the accepted WDW-6 physical quotient onto the no-spectator generated fibres  $\mathcal{L}_\tau^C$ . The maps  $U^C$  are strongly continuous and unitary and satisfy*

$$U^C(\tau, \tau) = I, \quad U^C(\tau_3, \tau_2)U^C(\tau_2, \tau_1) = U^C(\tau_3, \tau_1), \quad U^C(\tau, \sigma)\text{Cond}_\sigma = \text{Cond}_\tau. \quad (30)$$

The companion connection eq. (28) restricts to a metric-compatible Hermitian connection on the generated line. For every differentiable section  $\xi_\tau = \alpha(\tau)Y_\tau^{(0)}$ ,

$$\nabla_\tau^C \xi_\tau = \dot{\alpha}(\tau)Y_\tau^{(0)}. \quad (31)$$

Hence

$$i\hbar \nabla_\tau^C \xi_\tau = H_{\text{rel}}^C(\tau)\xi_\tau \iff \dot{\alpha}(\tau) = 0 \iff \xi_\tau = U^C(\tau, \sigma)\xi_\sigma. \quad (32)$$

The zero Hamiltonian is bounded and self-adjoint on the full fibre. Most importantly, the complete implication is constraint induced:

$$\widehat{\mathcal{C}}^{\text{ren}} F_{A,\mu}^{(0)} = 0 \implies \dot{Y}_\tau^{(0)} = \mathbb{A}_\tau Y_\tau^{(0)} \implies i\hbar \nabla_\tau^C Y_\tau^{(0)} = H_{\text{rel}}^C(\tau)Y_\tau^{(0)}. \quad (33)$$

*Proof.* Theorem 12.1 makes  $Y_\tau^{(0)}$  smooth, and nonvanishing at the anchor makes eq. (23) a nonempty open interval. The canonical coordinate  $I_6$  and eq. (26) make every  $\text{Cond}_\tau$  unitary onto its range. The intertwining, identity, composition, strong continuity, and unitarity laws follow directly from eq. (27).

Theorem 14.1 gives  $\dot{Y}_\tau^{(0)} = \mathbb{A}_\tau Y_\tau^{(0)}$  before the connection is defined. Substitution into  $\xi_\tau = \alpha(\tau)Y_\tau^{(0)}$  proves eq. (31); in particular the ambient companion connection maps differentiable generated-line sections back into the same fibre. Differentiating the pushed-forward form and using eq. (31) proves metric compatibility. Equation eq. (32) follows because  $H_{\text{rel}}^C = 0$ . The self-adjointness statement is immediate on the one-dimensional full domain.  $\square$

**Audit finding 15.2** (Why this is neither point relabeling nor a cancelling connection). The conditioning map is an explicit isometry from the accepted WDW-6 quotient into the actual value/normal-response germ doublet. It is therefore not the point character renamed. The physical clock was fixed independently by W7.1; the analytic splitting retains the mean-zero profile germ and contributes no clock selector. The operator  $\mathbb{A}_\tau$  is obtained by reapplying the renormalized constraint in W7.4, and only then is  $\nabla^C = \partial_\tau - \mathbb{A}_\tau$  restricted to the generated line. No auxiliary  $K$  is selected and no formula  $\Gamma = K - \mathbb{A}$  is used. The zero covariant Hamiltonian records parallel transport by the actual constraint-induced companion response. It is not an assertion that  $\mathbb{A}_\tau$  vanishes or is skew-adjoint on the enlarged full-Cauchy carrier.

## 16. Why the Cauchy doublet is minimal

Analytic restriction alone is not injective on second-order solution germs: two germs can agree at  $T = \tau$  and have different normal derivative. The scale-cylinder result exhibits the classical analogue exactly, while the phase-jet theorem proves first-phase-jet sufficiency on the complete regular history sector [14, 15]. The doublet is therefore the first non-lossy local trace compatible with the differential order of the WDW constraint.

**Proposition 16.1** (Minimal differential order). *Any local linear analytic Cauchy-germ carrier that factors the derivative of every second-order WDW solution germ must distinguish both  $R_\tau F$  and  $P_\tau F$ . Conversely, W7.4 shows that no higher  $T$ -normal derivative is required.*

*Proof.* If the carrier identifies germs with equal restriction but unequal first normal derivative, their first clock response differs, so derivative factorization fails. Necessity follows. Sufficiency follows because the WDW equation expresses the second normal derivative using only the two retained components and tangential derivatives of the first.  $\square$

**Theorem 16.2** (Source-generated conditioning exhaustion). *On the frozen WDW-6 handoff and physical profile-clock lineage, the source-generated no-spectator conditioning class has one lawful trace type. A point-only trace fails first-normal-response factorization; a one-component positive-frequency trace imports an unproved polarization; a scalar-functional clock fails W7.1; and an enriched response fibre changes the comparison class. The surviving same-scope carrier is the Cauchy-germ doublet conditioned through  $I_6$ , up to fibre frame and analytic-splitting skin.*

*Proof.* The preceding proposition excludes every lower differential-order trace. The arc contract forbids a positive-frequency selector without an independently proved polarization. Theorem 8.2 excludes scalar-functional clock substitutions. The WDW-6 quotient is rank one, so factorization through  $I_6$  and the no-spectator condition force the range to be  $\mathbb{C}J_\tau F_{A,\mu}^{(0)}$ . A larger positive form, additional conditional state, clock POVM, or full response completion introduces standing-bearing carrier data and is therefore an openly enriched scope, not a rival inside this class. Fixed-domain exhaustion and quotient rigidity leave only frame, complement, and coordinate changes that preserve the complete Cauchy germ, all of which are bookkeeping-equivalent [25, 26].  $\square$

The primary generated line contains the complete Cauchy germ doublet of the accepted physical state, but it does not claim to populate every arbitrary second-order solution germ. The larger response construction below addresses that separate carrier problem.

## Part IV: Physical algebra, complete records, and residual constraints

### 17. Fixed physical algebra and source-level observables

The inherited WDW-6 represented physical algebra is

$$\mathfrak{A}_6^{\text{pt}} = \mathbb{C}I.$$

Its action on the generated Cauchy line is

$$\rho_\tau^{\text{C}}(zI)(\alpha Y_\tau^{(0)}) = z\alpha Y_\tau^{(0)}. \quad (34)$$

This fixed algebra is intertwined by the constraint-induced propagator:

$$U^{\text{C}}(\tau, \sigma)\rho_\sigma^{\text{C}}(zI) = \rho_\tau^{\text{C}}(zI)U^{\text{C}}(\tau, \sigma). \quad (35)$$

No varying scalar  $O(x_\tau)$  is inserted into this identity.

Let  $\mathfrak{A}_{\text{cyl}}^{\text{inv}}$  be the WDW-2 invariant cylinder/local algebra represented on the complete WDW-5/6 graph source. For  $O$  in this algebra, let

$$O_\tau^{\text{src}} = R_\tau O \quad (36)$$

denote its directly descended slice action on both Cauchy components. A first normal jet gives the corresponding  $2 \times 2$  residual differential matrix whenever momentum content is present. This definition precedes every Hilbert completion.

## 18. Complete volume record on the occupied constraint orbit

Let  $\Gamma_*$  denote the occupied WDW-4A classical constraint orbit, including all proper spatial redescrptions and lapse/refoliation presentations licensed by WDW-3. On  $I_7^C$ , the clock guards make the function-valued scalar gauge  $\phi(\cdot) = \tau \mathbf{1}$  transverse and give one orbit point modulo proper spatial redescription:

$$s_\tau = [\Gamma_* \cap \{\phi(\cdot) = \tau \mathbf{1}\}]_{\text{pr}}. \quad (37)$$

Define the complete spatial-volume record

$$\mathcal{V}^{\text{rec}}(\tau) = \text{Vol}(h(s_\tau)). \quad (38)$$

This is a clock-indexed complete record on one constrained orbit. It is not represented as one fixed operator whose unequal scalar values are required to intertwine across time.

**Theorem 18.1** (W7.6: physical algebra, source observable, and complete-record descent). *The family  $\mathfrak{A}_{7,\tau}^{\text{src}}$  generated by restrictions and first normal jets of invariant WDW-2 observables:*

1. *is independent of connected proper presentation;*
2. *preserves the finite-cap algebraic response grammar inherited from the serialized source;*
3. *has its Hilbert boundedness, closability, adjoint, and graph-domain status decided operator by operator; and*
4. *separates every distinction detected by the admitted analytic cylinder/local subgrammar.*

*The fixed physical algebra eq. (34) is a bounded unital star representation on every generated fibre and satisfies eq. (35). The complete record eq. (38) is independent of connected proper spatial presentation, lapse parametrization, and refoliation presentation of the same occupied orbit, and it is compatible with scale identity transport.*

*Proof.* Proper invariance of  $O$  and of the quotient slice gives  $O_\tau^{\text{src}}(g_\tau y) = O_\tau^{\text{src}}(y)$ . Multiplication, first normal differentiation, and every residual differential word act capwise in the complete serialized AST, so they preserve the finite-cap source grammar. No Hilbert conclusion is drawn here: W7.7–9 later construct the form and decide the actual adjoint and evaluation domains. No finite matrix truncation or automatic bounded extension is used for the Cauchy source. Analytic cylinder polynomials separate the admitted residual source distinctions and their first jets separate its tangent directions.

Equation eq. (35) follows because the scalar  $z$  is the same represented algebra element at both endpoints. For the record, WDW-4A proves that the timelike scalar gradient makes  $\phi = \tau$  a geometric spacetime level set and that the profile gauge is transverse to the complete normal characteristic distribution. The clock guards therefore give a unique intersection with  $\Gamma_*$  on the certified interval. Proper spatial diffeomorphisms preserve the induced total volume. Changing lapse or foliation presentation while retaining the same constraint orbit and the same scalar level set changes only the parameterization/presentation of the intersection. Therefore eq. (38) factors through the clock-use and proper-redescription quotients. The auxiliary mean coordinate is absent from this descent argument. The inherited scale maps fix the orbit, scalar readout, and metric data literally.  $\square$

The inherited algebra and the source observable grammar have different scopes. The former remains  $\mathbb{C}I$  on the generated physical line; the latter is a downstream residual object and is not obtained by renaming that scalar algebra.

## 19. Residual constraint surface

Use the constant lapse mode  $N_0 = \mathbf{1}$  for the analytic normal decomposition. Since  $\mathcal{L}_X N_0 = 0$ , WDW-3 closure makes its Hamiltonian response preserve the proper-constraint kernel. This use does not identify lapse with the physical clock; the latter remains  $\phi(\cdot) = \tau \mathbf{1}$ . Every residual Hamiltonian profile and every

momentum constraint induces a compatibility operator  $\mathcal{R}_{\alpha,\tau}$  on the Cauchy-germ core. After W7.9–10 construct the history carrier and absolute evaluation domain, define

$$\mathcal{D}_{s,\tau}^{\text{phys}} = \left\{ \Xi \in \mathcal{D}_{\text{ev},s}^1 : \mathcal{R}_{\alpha,\tau} \text{Ev}_1 \Xi = 0 \text{ for every } \alpha \right\}. \quad (39)$$

**Lemma 19.1** (Constraint covariance on the common Cauchy domain). *The actual authenticated Cauchy curve satisfies every residual constraint by direct restriction of the joint-kernel identities. On every history datum lying in an  $\mathcal{E}$ -invariant absolute evaluation domain and having vanishing evaluated constraints, the history companion evolution preserves that joint residual kernel until the first anomaly, domain, or radius failure. Membership of the authenticated history Dyson series in that absolute domain is not asserted.*

*Proof.* The first statement follows by restricting the complete joint annihilation identities for  $F_{A,\mu}^{(0)}$ . For the second, absolute evaluation permits termwise application of the residual operators. Differentiate every residual identity along the companion evolution. Commuting its clock response with a registered residual constraint produces only the complete WDW-6 constraint combination; the anomaly remainder is zero on the accepted baseline. Uniqueness of the common-domain evolution in theorem 24.1 then keeps the zero solution zero.  $\square$

## 20. Nonconstant complete-record witness

The WDW-4A occupied history gives

$$\frac{d}{d\tau} \mathcal{V}^{\text{rec}}(0) = \frac{3c_{\epsilon_*}}{\epsilon_*} \text{Vol}(h_*) > 0. \quad (40)$$

Indeed,  $d \text{Vol} / dt = 3c_{\epsilon_*} \text{Vol}(h_*)$  while  $d\tau/dt = \epsilon_*$  at the anchor. Their ratio is invariant under lapse reparametrization. The complete record is therefore nonconstant on every sufficiently small interval containing the anchor. It is not used to construct the conditional form or to select a polarization.

# Part V: Compatible forms, genuine star envelopes, and companion evolution

The state-independent companion operator used throughout this part is the constraint-derived operator of W7.4. No form, adjoint, completion, or Hamiltonian is imported with it.

## 21. Frozen increment coordinates and the grade-zero form

Let  $\pi_N$  be the  $N$ th saturated finite Cauchy-response coordinate map on the exact algebraic source and define

$$Q_N := \text{im } \pi_N. \quad (41)$$

Thus  $Q_N$  is the finite-dimensional space of cap data that actually possess a source lift, not an arbitrarily enlarged formal coefficient space. The restriction  $r_{N+1,N} : Q_{N+1} \rightarrow Q_N$  is surjective: if  $q = \pi_N x$ , then  $q = r_{N+1,N} \pi_{N+1} x$ . Compute the reduced-row-echelon matrix of  $r_{N+1,N}$  in the frozen ordered bases and set every nonpivot coordinate to zero. This gives an explicit linear section

$$s_{N,N+1} : Q_N \longrightarrow Q_{N+1}, \quad r_{N+1,N} s_{N,N+1} = I. \quad (42)$$

This is a local finite-dimensional construction. The finite-anticipation result is used only to make cap coordinates and finite dependency queries effective; it is not credited with a Hilbert form or a general splitting theorem. Hence

$$Q_{N+1} = s_{N,N+1} Q_N \oplus \ker r_{N+1,N}$$

in the frozen echelon coordinates. Composing the sections gives coherent embeddings of earlier caps into later ones and constructs the algebraic coefficient module

$$\mathcal{V}_{\text{cap}} := \varinjlim (Q_N, s_{N,N+1}). \quad (43)$$

This direct limit is a newly constructed response-coordinate module; its basis vectors are not asserted individually to be global WDW sources. The coordinates on the second summand are the new increments and vanish on the selected old-cap extension. Enumerate the resulting increments in cap order as

$$\lambda_1, \lambda_2, \dots : \mathcal{V}_{\text{cap}} \longrightarrow \mathbb{C}. \quad (44)$$

The same symbols denote the compatible coefficient functionals on the exact source image. They are response coefficients, not evaluations at configuration points. By definition of the exact source quotient and exhaustion by the saturated cap maps,

$$x = 0 \text{ in the source} \iff \lambda_n(x) = 0 \text{ for every } n. \quad (45)$$

Let  $e_n \in \mathcal{V}_{\text{cap}}$  be the direct-limit coefficient vector whose  $n$ th increment is one and whose other increments are zero. Finite anticipation expresses each output increment of a serialized AST letter as a finite linear combination of input increments. Those coefficients define a row-finite matrix on the increment module; cap locality makes its columns finite. The matrix agrees with the exact source action on every embedded exact source sequence, but its definition on  $\mathcal{V}_{\text{cap}}$  does not require each  $e_n$  to have a global source lift or the selected cap sections to intertwine the AST.

Start with the countable family  $(e_n)$ , add the coefficient sequences of the authenticated Cauchy datum and all its finite forward responses, and saturate this family under finite *forward* AST words. Enumerate the resulting coefficient family as  $(z_j)_{j \geq 1}$ , with every  $e_n$  included. No dagger is used in this enumeration. Define

$$d_n = \frac{2^{-n}}{1 + \sum_{j=1}^n |\lambda_n(z_j)|^2}, \quad d_n > 0, \quad (46)$$

and, first on  $\mathcal{V}_{\text{cap}}$  and then on every exact source sequence for which the sum is finite,

$$b_0(x, y) = \sum_{n \geq 1} d_n \overline{\lambda_n(x)} \lambda_n(y). \quad (47)$$

For cap  $N$ , the sum stops after the last increment belonging to that cap; call the result  $b_0^{(N)}$ . Compatibility means

$$b_0^{(N+1)}(s_{N,N+1}x, s_{N,N+1}y) = b_0^{(N)}(x, y), \quad (48)$$

not pullback along an arbitrary formal cap.

**Theorem 21.1** (W7.7: explicit compatible grade-zero positive form). *The forms  $b_0^{(N)}$  are positive and compatible under the explicit sections eq. (42). On the exact source image their common radical is exactly the capwise zero-source relation. The full form  $b_0$  is positive definite on the finite-norm coordinate quotient. Every  $z_j$ , in particular the embedded authenticated datum  $Y_0^{(0)} = J_0 F_{A,\mu}^{(0)}$  and all of its finite forward AST responses, has finite  $b_0$  norm. Explicitly,*

$$\mathcal{H}_0^{\text{src}} = \ell^2(\mathbb{N}, d) = \overline{\mathcal{V}_{\text{cap}}}^{b_0} \quad (49)$$

is a separable Hilbert space containing the dense finite-cap coordinate core  $\mathcal{D}_{\text{cap}} := \mathcal{V}_{\text{cap}}$ . Exact source sequences enter this carrier precisely when their displayed weighted coordinate sum is finite.

*Proof.* Positivity is immediate from  $d_n > 0$ . The increment coordinates added at cap  $N + 1$  vanish on the section image, so eq. (48) holds. Equation eq. (45) identifies the radical exactly.

For fixed  $j$  and  $n \geq j$ ,

$$d_n |\lambda_n(z_j)|^2 \leq 2^{-n}.$$

The finitely many terms  $n < j$  are finite. Hence  $b_0(z_j, z_j) < \infty$ . Because every increment vector  $e_n$  occurs among the  $z_j$ , finite increment combinations form a countable-coordinate dense core after completion; hence the completion is separable and  $\mathcal{D}_{\text{cap}}$  is dense. The recursion is source-labelled and includes the whole finite-cap core before propagation; it is not a sampled configuration metric or an orbit-fitted norm.  $\square$

## 22. Actual source adjoints

The direct-limit vectors  $e_n$  satisfy  $b_0(e_n, e_m) = d_n \delta_{nm}$ . For a registered forward source letter  $L$ , let the same symbol denote its coefficient action constructed above and write

$$Le_n = \sum_m a_{mn}^L e_m. \quad (50)$$

Cap locality says that a finite input cap is sent into one finite enlarged cap, so every column of  $a^L$  is finite. Finite anticipation says that each output increment depends on finitely many input increments, so every row is finite. On embedded exact sources this matrix is the registered source action; on the full coordinate core it is its explicitly constructed extension. On  $\mathcal{D}_{\text{cap}}$  define

$$L^\dagger e_m = \sum_n \frac{d_m}{d_n} \overline{a_{mn}^L} e_n. \quad (51)$$

The sum is finite by row finiteness.

The maximal Hilbert adjoint is

$$\text{Dom } L^* = \left\{ y = (y_m) : \left( \frac{1}{d_n} \sum_m d_m \overline{a_{mn}^L} y_m \right)_n \in \ell^2(d) \right\}, \quad (52)$$

$$(L^* y)_n = \frac{1}{d_n} \sum_m d_m \overline{a_{mn}^L} y_m. \quad (53)$$

**Theorem 22.1** (W7.8: genuine source star representation). *Every forward AST letter is densely defined and closable on  $\mathcal{H}_0^{\text{src}}$ . Its registered dagger is not an independent formal channel: it is the restriction*

$$L^\dagger = L^*|_{\mathcal{D}_{\text{cap}}}, \quad (54)$$

with  $L^*$  given by eqs. (52) and (53). The finite-cap core is invariant under  $L$  and  $L^\dagger$ , and

$$b_0(Lx, y) = b_0(x, L^\dagger y), \quad (L^\dagger)^\dagger x = Lx \quad (x, y \in \mathcal{D}_{\text{cap}}). \quad (55)$$

Thus the algebra generated by the forward letters and eq. (51) is a genuine dense-domain  $*$ -representation.

*Proof.* For basis vectors,

$$b_0(Le_n, e_m) = d_m \overline{a_{mn}^L} = b_0(e_n, L^\dagger e_m).$$

Finite linearity proves the first identity. Applying the weighted conjugate transpose twice returns  $a^L$ , proving involution. The displayed maximal formula is obtained by solving  $b_0(Lx, y) = b_0(x, z)$  for  $z$ ; hence it is the actual Hilbert adjoint, not merely a transpose label. Dense adjoint domain implies closability.  $\square$

## 23. Unquotiented response-history Fock envelope

Let  $\mathfrak{L}$  be the finite serialized *positive-response* alphabet of the companion block, with cost  $r(L) \in \{1, 2\}$ . The upper-right identity is a block transition and does not create a word letter. For a word  $W = L_k \cdots L_1$ , put  $q(W) = \sum_j r(L_j)$ . Choose a star-symmetric multiplicative history coefficient

$$M(W) = \prod_{j=1}^k m(L_j), \quad m(L) = m(L^\dagger) \geq 1, \quad (56)$$

for the finitely many companion-block symbols. After shrinking  $J_7$ , it records uniform bounds for their continuous time-dependent scalar and finite-matrix coefficients; it is not asserted to dominate every evaluated source word. For  $0 < s < s_*$  define the response-history carrier

$$\mathcal{F}_s = \bigoplus_{W \in \mathcal{Q}^*}^{\ell^2} (W \otimes \mathcal{H}_0^{\text{src}}), \quad \|W \otimes u\|_s^2 = e^{2sq(W)} q(W)! M(W)^2 \|u\|_0^2. \quad (57)$$

Distinct histories remain orthogonal even when their evaluated source responses agree. In particular, no evaluation radical is quotiented.

Let  $c_s(L)$  be left creation,  $c_s(L)(W \otimes u) = LW \otimes u$ . Its actual Hilbert adjoint is the prefix-removal operator

$$c_s(L)^*(L'W \otimes u) = \delta_{L,L'} e^{2sr(L)} m(L)^2 \frac{(q(W) + r(L))!}{q(W)!} W \otimes u, \quad (58)$$

on its maximal weighted domain. Formula eq. (58), not left creation by a second dagger letter, is the registered response dagger.

Let  $\mathcal{D}_{\text{src}}$  be the span of histories  $W \otimes u$  with  $u \in \mathcal{D}_{\text{cap}}$ . Define source evaluation there by

$$\text{Ev}(L_k \cdots L_1 \otimes u) = L_k \cdots L_1 u. \quad (59)$$

For a completed history  $\xi = (\xi_W)_W$ , define the absolutely source-majorized domain

$$\mathcal{D}_{\text{ev},s}^1 = \left\{ \xi : \xi_W \in \text{Dom } W \text{ and } \sum_W \|W \xi_W\|_{\mathcal{H}_0^{\text{src}}} < \infty \right\}, \quad (60)$$

and set

$$\text{Ev}_1 \xi = \sum_W W \xi_W. \quad (61)$$

Absolute convergence makes this a single-valued linear map. No assertion is made that  $\text{Ev}$  is closable on all of  $\mathcal{F}_s$ , or that  $\mathcal{D}_{\text{ev},s}^1$  contains every completed source history.

**Theorem 23.1** (W7.9: genuine response-history star envelope). *On the dense finite-history core  $\mathcal{D}_{\text{fin}}$ :*

1.  $c_s(L)^*$  is exactly eq. (58);
2.  $L \mapsto c_s(L)$  and  $L^* \mapsto c_s(L)^*$  define a genuine  $*$ -representation;
3. the algebraic evaluation map satisfies

$$\text{Ev } c_s(L) \xi = L \text{ Ev } \xi \quad (62)$$

on  $\mathcal{D}_{\text{src}}$ , and the same identity holds for  $\text{Ev}_1$  whenever both sides lie in the absolute domain; and

4. quotienting by  $\text{Ker Ev}$  is neither used nor needed.

*Proof.* Orthogonality of word sectors and the ratio of the two weights in eq. (57) give eq. (58). This proves the star identity directly. Equation eq. (62) is the definition of word evaluation and associativity of the actual source action. Absolute convergence permits regrouping on  $\mathcal{D}_{\text{ev},s}^1$ . No reducing-kernel or unrestricted closability inference is used.  $\square$

## 24. Graded companion evolution

Lift the companion block to the history carrier by replacing each response letter with its left-creation action; retain the upper-right block identity. Denote the result by  $\mathcal{A}_\tau$ . Then

$$\text{Ev } \mathcal{A}_\tau \xi = \mathbb{A}_\tau \text{ Ev } \xi \quad (63)$$

on the algebraic source-history core, and for  $\text{Ev}_1$  on the common absolute domain.

For a matrix path  $p$  through a product of  $n$  companion blocks, let  $N_C(p)$ ,  $N_D(p)$ , and  $N_I(p)$  count the lower-left, lower-right, and upper-right legs. Its response cost is

$$q(p) = 2N_C(p) + N_D(p). \quad (64)$$

Every  $C$  after the first requires an intervening upper-right leg, so

$$\boxed{q(p) \leq n + 1.} \quad (65)$$

For  $0 < s' < s < s_*$ ,  $\delta = s - s' \leq 1$ , there are source constants  $B, C > 0$  such that

$$\|W_p(\tau_n, \dots, \tau_1)\xi\|_{s'} \leq CB^n \delta^{-(n+1)/2} \sqrt{(n+1)!} \|\xi\|_s. \quad (66)$$

Indeed, a cost- $q$  creation word has grade factor  $\sqrt{(k+q)!/k!}$ , and

$$\sup_{k \geq 0} e^{-\delta k} \sqrt{(k+q)!/k!} \leq C_1^q \delta^{-q/2} \sqrt{q!}.$$

Let

$$\mathcal{E}_n(\tau, \sigma)\xi = \int_{\sigma < t_n < \dots < t_1 < \tau} \mathcal{A}_{t_1} \cdots \mathcal{A}_{t_n} \xi dt_n \cdots dt_1, \quad \mathcal{E}_0 = I. \quad (67)$$

With  $T = |\tau - \sigma|$  and  $z = 2BT/\sqrt{\delta}$ , the even and odd bounds are

$$e_m(\delta, T) = \frac{2C}{\sqrt{\delta}} \frac{z^{2m} \sqrt{(2m+1)!}}{(2m)!}, \quad (68)$$

$$o_m(\delta, T) = \frac{2C}{\sqrt{\delta}} \frac{z^{2m+1} \sqrt{(2m+2)!}}{(2m+1)!}. \quad (69)$$

Their ratios tend to zero:

$$\frac{e_{m+1}}{e_m} = z^2 \frac{\sqrt{(2m+3)(2m+2)}}{(2m+2)(2m+1)} \rightarrow 0, \quad (70)$$

$$\frac{o_{m+1}}{o_m} = z^2 \frac{\sqrt{(2m+4)(2m+3)}}{(2m+3)(2m+2)} \rightarrow 0. \quad (71)$$

**Theorem 24.1** (W7.10: history evolution and evaluated companion response). *The series*

$$\mathcal{E}(\tau, \sigma) = \sum_{n \geq 0} \mathcal{E}_n(\tau, \sigma) \quad (72)$$

*converges absolutely in  $\mathcal{B}(\mathcal{F}_s, \mathcal{F}_{s'})$ , locally uniformly on  $J_T^2$ . It has the strong-scale evolution, identity, composition, and reverse-time properties. If a datum belongs to an  $\mathcal{E}$ -invariant absolute source-majorized sector, then*

$$\text{Ev}_1 \mathcal{E}(\tau, \sigma) \iota J_\sigma F = J_\tau F \quad (73)$$

*where  $\iota$  is the empty-history embedding. The theorem does not assert that every joint solution, or the authenticated one, satisfies this additional absolute source-growth condition.*

*Proof.* The norms of the even and odd Dyson terms are bounded by eqs. (68) and (69); eqs. (70) and (71) give convergence. An additional factor gives termwise differentiation in a weaker radius. Volterra uniqueness gives composition and inversion. On an invariant absolute source-majorized sector, eq. (63) permits termwise evaluation, and W7.4 plus uniqueness in that sector gives eq. (73). No finite cap is declared invariant and no relation quotient is applied to the history Hilbert space.  $\square$

## 25. Carleman closure in the genuine star representation

At the anchor define the response-history skew operator

$$\mathcal{K}_0^{\text{pre}} = \frac{1}{2}(\mathcal{A}_0 - \mathcal{A}_0^*) \quad \text{on } \mathcal{D}_{\text{fin}}, \quad \mathcal{K}_0^* = \overline{i\hbar\mathcal{K}_0^{\text{pre}}}. \quad (74)$$

Here  $\mathcal{A}_0^*$  is the actual Hilbert adjoint determined by eq. (58).

**Theorem 25.1** (W7.11: genuine Carleman closure). *For every finite-history vector  $\xi$  there are  $C_\xi, R_\xi < \infty$  such that*

$$\|(\mathcal{K}_0^{\text{pre}})^m \xi\|_s \leq C_\xi R_\xi^m \sqrt{(2m)!}, \quad m \geq 0. \quad (75)$$

Therefore

$$\sum_{m \geq 1} \|(\mathcal{K}_0^{\text{pre}})^m \xi\|_s^{-1/m} = \infty. \quad (76)$$

The symmetric operator  $i\hbar\mathcal{K}_0^{\text{pre}}$  is essentially self-adjoint, so  $\mathcal{K}_0^*$  is self-adjoint. The empty-history authenticated vector  $\iota Y_0^{(0)}$  lies in  $\text{Dom } \mathcal{K}_0^*$ .

*Proof.* The operator is skew-symmetric because  $\mathcal{A}_0^*$  is the actual adjoint. A product of  $m$  creation or annihilation blocks changes response cost by at most  $2m$ . Starting from top cost  $N_\xi$ , the factorial weight gives

$$\|W\xi\|_s \leq C_\xi b^m \sqrt{\frac{(N_\xi + 2m)!}{N_\xi!}} \leq C_\xi R_\xi^m \sqrt{(2m)!}.$$

The last fixed-degree polynomial is absorbed into  $R_\xi^m$ . Expansion of the  $m$ th power has  $2^m$  terms and the prefactor  $2^{-m}$ . Stirling yields the harmonic Carleman divergence. The finite-history core is total, so Nussbaum's quasi-analytic-vector theorem gives essential self-adjointness of  $i\hbar\mathcal{K}_0^{\text{pre}}$  [29].

For occupancy, finite-cap approximants to  $Y_0^{(0)}$  converge in  $\mathcal{H}_0^{\text{src}}$ . On empty histories, one application of  $\mathcal{K}_0^{\text{pre}}$  has only finitely many cost-at-most-two output sectors and is continuous in the grade-zero vector. The approximants and their images therefore converge, and the graph-closure criterion places  $\iota Y_0^{(0)}$  in the closed domain.  $\square$

## 26. Transported Hilbert fibres and the enlarged-carrier Hamiltonian boundary

Choose radii  $s_+ > s$  and a time interval on which W7.10 maps the dense positive-reserve domain  $\mathcal{D}_+ := \mathcal{F}_{s_+}$  injectively into  $\mathcal{F}_s$  and the reverse evolution is defined after inserting an intermediate radius. Put  $\mathcal{S}_\tau = \mathcal{E}(\tau, 0)|_{\mathcal{D}_+}$  and define on its range

$$B_\tau(\mathcal{S}_\tau X, \mathcal{S}_\tau Y) = \langle X, Y \rangle_{\mathcal{F}_s}. \quad (77)$$

Let  $\mathcal{H}_\tau^{\text{comp}}$  be the corresponding fibre completion. The pullback definition makes  $\mathcal{S}_\tau$  unitary from the completion of  $\mathcal{D}_+$  in the anchor norm onto  $\mathcal{H}_\tau^{\text{comp}}$ , and composition from W7.10 makes  $\mathcal{E}(\tau, \sigma)$  unitary between these fibres. This is a Hilbert bundle built from the positive-reserve analytic scale; it is not a claim that  $\mathcal{E}(\tau, \sigma)$  is bounded and invertible on one fixed-radius space.

The companion transport connection is

$$\nabla_\tau^{\text{comp}} = \partial_\tau - \mathcal{A}_\tau. \quad (78)$$

It is metric compatible with eq. (77), and history companion solutions are parallel:

$$\nabla_\tau^{\text{comp}} \Xi_\tau = 0. \quad (79)$$

**Theorem 26.1** (W7.12: full-Cauchy Hilbert-bundle extension and non-cancelling Hamiltonian gate). *The data through W7.11 construct a state-independent response-history Hilbert-bundle companion evolution with a positive grade-zero source form, genuine star representation, algebraic and absolutely source-majorized evaluation intertwiners, and Carleman-closed skew star envelope. They do not prove unrestricted completed evaluation or a nonzero relational Hamiltonian generating the actual full-Cauchy companion derivative. This limitation does not reopen the source-generated endpoint proved in W7.5: the two theorems act on different carriers and only the generated line is already tied to the authenticated evaluated Cauchy curve.*

More precisely, an equation

$$i\hbar(\partial_\tau + \Gamma_\tau)\Xi_\tau = H_\tau\Xi_\tau \quad (80)$$

is equivalent on all companion solutions to  $\partial_\tau\Xi_\tau = \mathcal{A}_\tau\Xi_\tau$  only if

$$H_\tau = i\hbar(\mathcal{A}_\tau + \Gamma_\tau) \quad (81)$$

on the common domain. Hence choosing

$$\Gamma_\tau = \mathcal{K}_\tau - \mathcal{A}_\tau, \quad H_\tau = i\hbar\mathcal{K}_\tau \quad (82)$$

is an algebraic cancellation, not a derivation of  $H_\tau$ . The canonical companion connection eq. (78) gives  $H_\tau = 0$  in eq. (81); on this enlarged history carrier that identity records formal history transport but is not the generated-line Hamiltonian of eq. (29), because unrestricted evaluation and authenticated absolute-domain occupancy are not proved here. With the fixed trivial connection  $\Gamma_\tau = 0$ , a nonzero full-Cauchy Hamiltonian would require  $\mathcal{A}_\tau$  itself to be skew-adjoint in one fixed Hilbert form, which is not proved and is not implied by the star-envelope theorem.

Therefore the exact enlarged-carrier status is

$$\boxed{\text{FULL-CAUCHY-COMPANION}}, \quad \text{NONZERO-HAMILTONIAN} = \text{OPEN} \quad (\text{full-Cauchy carrier}). \quad (83)$$

*Proof.* Metric compatibility follows by differentiating eq. (77) and using  $\dot{\mathcal{S}}_\tau = \mathcal{A}_\tau\mathcal{S}_\tau$ . Equation eq. (79) follows directly. Substituting a companion solution into eq. (80) gives eq. (81); conversely that identity recovers the companion equation. Formula eq. (82) therefore has no independent selecting content. Carleman closure proves lawfulness of  $\mathcal{H}_0^*$  as a self-adjoint operator in the response envelope, but no preceding theorem identifies  $\mathcal{H}_0^*$  with  $i\hbar\mathcal{A}_0$  or supplies an independently fixed nontrivial connection. This proves only the stated boundary for the stronger full-Cauchy extension; W7.5 already has the constraint-restricted connection and full-domain generator on the narrower source-generated line.  $\square$

## 26.1 Exact metric-polar extension criterion

The following calculation isolates the additional theorem needed for a Hamiltonian endpoint. Suppose there exists an evaluated physical response sector  $\mathcal{H}_\tau^{\text{ev}}$  containing  $J_\tau F_{A,\mu}^{(0)}$ , together with an invertible evolution  $E^{\text{ev}}(\tau, \sigma)$  such that

$$E^{\text{ev}}(\tau, \sigma)J_\sigma F = J_\tau F \quad (F \in \mathcal{Z}_{A,\mu}) \quad (84)$$

on a stable common domain. Transporting a positive anchor form gives

$$G_\tau = E^{\text{ev}}(\tau, 0)^{-*}G_0E^{\text{ev}}(\tau, 0)^{-1}, \quad \dot{G}_\tau = -\mathbb{A}_\tau^*G_\tau - G_\tau\mathbb{A}_\tau. \quad (85)$$

Relative to a fixed source trivialization, the unique metric-compatible connection whose connection form is  $G_\tau$ -self-adjoint would then be

$$\Gamma_\tau^{\text{min}} = \frac{1}{2}G_\tau^{-1}\dot{G}_\tau = -\frac{1}{2}(\mathbb{A}_\tau + \mathbb{A}_\tau^{\sharp_{G_\tau}}). \quad (86)$$

Consequently the companion equation would take the non-cancelling covariant Schrödinger form

$$i\hbar(\partial_\tau + \Gamma_\tau^{\text{min}})Y_\tau = H_\tau^{\text{rel}}Y_\tau, \quad H_\tau^{\text{rel}} = \frac{i\hbar}{2}(\mathbb{A}_\tau - \mathbb{A}_\tau^{\sharp_{G_\tau}}). \quad (87)$$

This implication is not used in the main theorem. W7.10 does not prove that the authenticated history Dyson series lies in the absolute evaluation domain, and finite anticipation alone gives no uniform analytic or Gevrey bound on repeated source responses. Thus  $E^{\text{ev}}$  in eq. (84) is not constructed in this theorem. Moreover, essential self-adjointness of the skew part in eq. (87) must be proved on the evaluated domain; it does not follow from Carleman closure of the free history envelope.

There is a second, independent covariance requirement. The zero-skew characterization in eq. (86) is unique only relative to the chosen source trivialization. Under a time-dependent unitary change of source frame, the connection acquires the frame's skew Maurer–Cartan term. Since the flow-box and response frames are presentation data in the equivalence relation below, a physical Hamiltonian theorem must either construct a canonical source trivialization or prove that the resulting Hamiltonian packages are equivalent under the full frame change. Neither statement is supplied by metric transport alone.

## 27. Direct relational-observable covariance

For a directly descended readout  $O_\tau^{\text{src}}$  preserving the evaluation and companion domains, let  $\mathcal{O}_\tau$  be its history lift and put

$$\tilde{\mathcal{O}}_\tau = \mathcal{S}_\tau^{-1} \mathcal{O}_\tau \mathcal{S}_\tau. \quad (88)$$

**Theorem 27.1** (W7.13: companion-covariant observables). *On the common domain,*

$$\partial_\tau \mathcal{O}_\tau - [\mathcal{A}_\tau, \mathcal{O}_\tau] = \mathcal{S}_\tau \partial_\tau \tilde{\mathcal{O}}_\tau \mathcal{S}_\tau^{-1}, \quad (89)$$

and

$$B_\tau(\Xi_\tau, \mathcal{O}_\tau \Xi_\tau) = B_0(\Xi_0, \tilde{\mathcal{O}}_\tau \Xi_0). \quad (90)$$

Evaluation of eq. (89) gives the direct source-response covariance whenever all terms lie in  $\mathcal{D}_{\text{ev},s}^1$ .

*Proof.* Differentiate eq. (88) and use  $\dot{\mathcal{S}}_\tau = \mathcal{A}_\tau \mathcal{S}_\tau$ . The expectation identity follows from fibre unitarity. No Hamiltonian commutator is inserted.  $\square$

## 28. Many-fingered compatibility and exact remaining gate

**Theorem 28.1** (W7.14: exact profile-time boundary). *The accepted WDW-6 commutator closure gives constraint-response flatness on the complete joint kernel. A state-independent operator-flat Tomonaga–Schwinger companion bundle additionally requires compatible grade-zero forms for every profile direction, one common star-envelope domain, factorization of all mixed profile Hessians there, and vanishing modulo residual constraints of the full companion-connection curvature. The single-clock construction proves none of those stronger statements.*

*Proof.* The graded estimates above use one constant-clock companion path. An operator curvature can annihilate the joint kernel while remaining nonzero on the response envelope. Mixed-profile form, domain, and curvature gates are therefore additional.  $\square$

# Part VI: Change, records, and failure

## 29. Generated-state transport and complete-record change

The theorem package distinguishes

$$\text{ResponseOp}_7 : \mathbb{A}_\tau \text{ is nonscalar on the common Cauchy domain,} \quad (91)$$

$$\text{RayChange}_7 : [Y_{\tau_2}^{(0)}] \neq [U^C(\tau_2, \tau_1)Y_{\tau_1}^{(0)}], \quad (92)$$

$$\text{RecordChange}_7 : \exists \tau_1, \tau_2 : \mathcal{V}^{\text{rec}}(\tau_1) \neq \mathcal{V}^{\text{rec}}(\tau_2), \quad (93)$$

$$\text{Change}_7 := \text{RayChange}_7 \vee \text{RecordChange}_7. \quad (94)$$

**Theorem 29.1** (W7.15: relational change and ray classification).  *$\mathbb{A}_\tau$  is nonscalar because its upper-right block is  $I$ . On every sufficiently small certified interval containing the anchor,*

$$\boxed{\text{RecordChange}_7 = \text{TRUE}}, \quad \boxed{\text{RayChange}_7 = \text{FALSE}}, \quad \boxed{\text{Change}_7 = \text{TRUE}}. \quad (95)$$

*The change witness is the complete record on the occupied constraint orbit, not a varying scalar falsely treated as one fixed intertwined quantum operator.*

*Proof.* The block argument proves nonscalarity of the state-independent Cauchy response. Equation eq. (40) gives a nonzero derivative of the complete volume record at the anchor, so the record is nonconstant on every sufficiently small interval. By eq. (27),  $U^C(\tau_2, \tau_1)$  maps  $Y_{\tau_1}^{(0)}$  to  $Y_{\tau_2}^{(0)}$ , so their transported projective classes agree. The record predicate and ray predicate are therefore distinct and have the stated values.  $\square$

## 30. Record factorization and no backflow

Let  $\rho_\tau$  contain the fixed source record, clock quotient, canonical WDW-6 coordinate, Cauchy conditioning map, generated line and pushed-forward form, constraint-derived companion restriction, generated propagator and connection, Hamiltonian graph, fixed scalar physical algebra, complete record, change predicates, compatible grade-zero weight recursion, source adjoint matrices and maximal domains, history-Fock recursion, even/odd evolution certificate, evaluation graph, Carleman closure, transported companion form, auxiliary companion propagator segment, endpoint label, and every pass/failure predicate through  $\tau$ . Two records are equivalent only when all standing-bearing coordinates match under the package intertwiners of Part VII.

**Theorem 30.1** (W7.16: record inheritance). *The base record quotient, conditioned verdict, retention order, and first-failure boundary are inherited. If  $\tau_1 < \tau_2$  in one lineage, later frame changes, smaller analytic radii, alternate compatible source weights, or a downstream enlarged-carrier Hamiltonian proposal cannot alter fixed coordinates at  $\tau_1$ .*

*Proof.* The record theorem of [27] applies because the domain, quotient, conditioning maps, complete readout, and continuation maps are explicit. Composition of  $U^C$  and  $\mathcal{E}$  makes temporal restriction functorial at their respective scopes. Changing the compatible source form produces a different companion-extension record unless a complete literal-source unitary exists. A later nonzero full-Cauchy Hamiltonian construction would extend the auxiliary record; it would not rewrite the source-generated endpoint. The standing-order specialization is [28].  $\square$

## 31. Quantum failure boundary

The source-generated lineage stops at the first loss of the WDW-4A profile-clock transversality or physical evaluation at  $\chi = 0$ , analytic quotient splitting, proper-redescription covariance, nonzero authenticated Cauchy-germ doublet, WDW-6 conditioning compatibility, no-spectator generation, companion restriction to the generated line, complete-record descent, or record continuity. The auxiliary full-Cauchy lineage additionally stops at the first:

1. zero of the clock-normal coefficient  $a_\tau$ ;
2. failure of finite-cap increment separation or compatibility;
3. divergence of the explicit grade-zero source form on a required source vector;
4. failure of cap locality or finite anticipation needed for the adjoint matrix;
5. nondensity of a maximal Hilbert-adjoint domain;
6. use of left creation by a dagger letter in place of prefix-removal adjunction;
7. failure of algebraic or absolute-domain evaluation intertwining;
8. divergence of either the even or odd history companion majorant;
9. failure of common-domain evolution, inverse, or composition;
10. failure of the genuine symmetry or Carleman condition for the star envelope;
11. exit of the authenticated datum from the star-envelope domain;
12. loss of positivity, coercivity, or transported companion-form compatibility;
13. promotion of the star envelope to a nonzero full-Cauchy Hamiltonian without a non-cancelling identification theorem;
14. loss of residual constraint propagation or appearance of an anomaly remainder;
15. failure of direct observable representation or covariance;
16. an unlicensed mixed-profile or holonomy promotion; or
17. loss of a record coordinate.

Failure of the enlarged-carrier Hamiltonian-identification gate is not a failure of either the profile-clock Cauchy-germ endpoint or the companion bundle. The exact physical label is **PROFILE-CLOCK-CAUCHY-GERM-WDW7-CLOSED**; the nonzero full-Cauchy Hamiltonian extension remains **OPEN**.

## Part VII: Generated realization quotient, companion extension, and capstone

### 32. The source-generated physical package

Define

$$\mathfrak{P}_7^{\text{gen}} = (\mathcal{D}_{\text{clk}}^{(7)}, \bar{\kappa}, \{\mathcal{S}_\tau^{\text{phys}}, \text{Ev}_\tau^\phi\}, I_7^C, \mathcal{H}_6^{\text{pt}}, I_6, \{\text{Cond}_\tau, \mathcal{L}_\tau^C, \langle \cdot, \cdot \rangle_\tau^C, Y_\tau^{(0)}\}, U^C, \nabla^C, H_{\text{rel}}^C, \{\rho_\tau^C\}, \mathcal{V}^{\text{rec}}, \mathcal{R}_7^{\text{gen}}, \mathcal{F}_7^{\text{gen}}, \mathcal{C}_7^{\text{gen}}). \quad (96)$$

The entries are, respectively, the inherited physical profile-clock package; auxiliary analytic splitting, physical  $\chi = 0$  slices, and evaluation maps; maximal generated-line interval; accepted WDW-6 physical quotient and canonical coordinate; conditioning maps, generated fibres, pushed-forward forms, and authenticated Cauchy-germ section; propagator, Hermitian companion connection, and full-domain self-adjoint generator; fixed scalar physical algebra; complete volume record; record quotient; first-failure object; and certificates. No residual measure, positive-frequency selector, full-source Hilbert form, or response-history completion is a coordinate of this primary package.

The realization class is defined by antecedents rather than by freezing the desired conclusion. Let  $\text{Real}_7^{\text{gen}}$  consist of packages satisfying all of the following:

1. the accepted WDW-4A function-valued clock-use package and WDW-6 quotient with canonical coordinate  $I_6$  are preserved;

2. the physical gauge is  $\phi(\cdot) = \tau \mathbf{1}$ ; any scalar mean or quotient flow-box coordinate is analytic skin and is not used to establish orbit transversality;
3. the clock history, proper quotient, authenticated joint solution, and complete Cauchy-germ trace are the frozen source objects;
4. the residual germ retains the mean-zero scalar profile and physical readout factors through  $\text{Ev}_\tau^\phi$  at  $\chi = 0$ ;
5. conditioning factors through  $I_6$  and the actual authenticated doublet  $Y_\tau^{(0)}$ ;
6. every conditional fibre is generated by the range of conditioning, with no spectator direction;
7. the conditional form is the pushforward of the WDW-6 physical form;
8. evolution is induced by the restriction of the state-independent companion equation to the generated fibres;
9. reported change is carried by a complete record at the unique  $\phi(\cdot) = \tau \mathbf{1}$  intersection of the occupied constraint orbit and descends through the clock/proper-redescription quotient; and
10. interval termination obeys the first-failure rule.

The definition does not prescribe a propagator formula, connection coefficient, Hamiltonian value, record value, or change verdict. Those are consequences of the ten requirements.

An equivalence contains an orientation-preserving WDW-4A clock reparametrization preserving the anchor and failure loci and fibre unitaries preserving the WDW-6 handoff, conditioning, generated fibres, forms, companion restriction, derived transport, fixed physical algebra, complete records, and record lineage. A change of analytic mean/complement chart that transports the complete Cauchy germ covariantly and a normalized phase-frame change are skin; promoting either to clock authority, adjoining a spectator fibre, changing the conditional form, or changing the complete-record orbit is not.

### 33. Finite source-generated candidate exhaustion

Every package in the frozen primary comparison class has three first-divergence coordinates: the proposed clock role, the trace order/polarization, and the carrier load. The preceding theorems give the following complete normal-form table.

| Candidate normal form                                                     | Status                      | First decisive gate                                                                                            |
|---------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------|
| External parameter                                                        | <b>ELIMINATED</b>           | No inherited clock-use quotient; violates the compact boundaryless constraint scope.                           |
| Lapse or foliation label                                                  | <b>ELIMINATED</b>           | WDW-3 presentation variable, not a clock record or standing-bearing quotient class.                            |
| Real-valued functional clock, including the scalar mean                   | <b>ELIMINATED</b>           | W7.1: its differential has a nonzero kernel on the full lapse distribution.                                    |
| Function-valued scalar clock with point-only trace                        | <b>ELIMINATED</b>           | W7.2–4: the normal response is lost and the second-order constraint does not factor.                           |
| Function-valued scalar clock with one frequency sector                    | <b>UNPOPULATED-SELECTOR</b> | No polarization, square-root domain, or extension theorem is inherited or constructed.                         |
| Function-valued scalar clock with Cauchy germ and no spectators           | <b>SURVIVOR</b>             | W7.1–6: profile transversality, full germ response, WDW-6 conditioning, and constraint factorization all pass. |
| Additional conditional states, positive forms, POVMs, or spectator fibres | <b>ENRICHED-SCOPE</b>       | Changes a standing-bearing carrier coordinate and leaves $\text{Real}_7^{\text{gen}}$ .                        |
| State-independent operator-flat profile bundle                            | <b>OPEN-EXTENSION</b>       | W7.14: mixed-profile forms, common domains, Hessians, and curvature remain unproved.                           |

**Proposition 33.1** (Exhaustiveness and lawful population). *The table is exhaustive for the frozen source-generated comparison class, and exactly one row is populated and passes every primary gate.*

*Proof.* The arc contract requires a clock role, a trace capable of factoring the accepted second-order constraint, and a declared physical carrier. WDW-3 and W7.1 exhaust the clock-role normal forms. The differential-order proposition and the conditioning-exhaustion theorem exhaust point, Cauchy, polarized, and enriched trace/carrier possibilities. Any extra distinction that changes neither the quotient/clock content nor the characteristic/constraint structure is skin; if it changes either, it is a different theorem object; if it carries force while changing neither, it is a forbidden second classifier or selector [25, 26]. W7.1–6 explicitly construct the surviving row, so the exhaustion is populated rather than a uniqueness statement over an empty class.  $\square$

**Theorem 33.2** (W7.17: complete source-generated endpoint). *The source-generated class is nonempty and*

$$\left[ |\text{Real}_7^{\text{gen}} / \sim_7| = 1, \quad H_{\text{rel}}^{\text{C}} = 0, \quad \text{RecordChange}_7 = \text{TRUE}, \quad \text{RayChange}_7 = \text{FALSE}. \right] \quad (97)$$

*This is the exact WDW-7 endpoint on the frozen profile-clock, source-generated, no-spectator comparison class. It is not a uniqueness theorem for clocks from different source lineages or for every conceivable enriched conditional carrier.*

*Proof.* W7.1 proves that the physical clock layer has one class and that scalar analytic coordinates carry no clock standing. W7.2–6 construct one package on that class. The conditioning exhaustion theorem eliminates every lower trace, hidden polarization, and same-scope spectator alternative. In any admitted package, factorization through the canonical coordinate and the no-spectator requirement therefore force

$$\text{Cond}_\tau[u] = I_6([u])Y_\tau^{(0)}, \quad \mathcal{L}_\tau^{\text{C}} = \mathbb{C}Y_\tau^{(0)}.$$

Pushforward of the WDW-6 form fixes the line metric. W7.4 fixes  $\dot{Y}_\tau^{(0)} = \mathbb{A}_\tau Y_\tau^{(0)}$ ; compatibility with conditioning therefore forces the unique coefficient-preserving transport  $U^{\text{C}}(\tau, \sigma)Y_\sigma^{(0)} = Y_\tau^{(0)}$  and the restriction  $\nabla^{\text{C}} = \partial_\tau - \mathbb{A}_\tau$ . On a generated rank-one fibre, the covariant Schrödinger equation then forces

the full-domain endomorphism  $H_{\text{rel}}^{\text{C}} = 0$ . A phase-frame change changes only the ordinary-coordinate representative and is removed by the declared equivalence.

W7.6 fixes the complete volume record from the occupied orbit and the function-valued clock intersection; W7.15 derives the change predicates, and W7.16 fixes records and failures. Analytic complement changes preserve the germ and are removed as skin. Thus every derived coordinate agrees under the unique fibre isometry, proving the singleton quotient without including those derived values in the class definition.  $\square$

**Audit finding 33.3 (Scope of the singleton).** The theorem exhausts only the inherited profile-clock, source-generated no-spectator class. A clock from another source or branch, a larger positive form, additional conditional states, non-scalar physical algebra, clock POVM, positive-frequency sector, or full-Cauchy Hamiltonian changes a standing-bearing coordinate and belongs to a different comparison class. No claim is made that such enriched classes are empty or unique.

## 34. Lawful full-Cauchy companion package

For an admitted analytic radius and a compatible source weight sequence  $d = (d_n)$  satisfying W7.7, define

$$\begin{aligned} \mathfrak{C}_7(d) = & (I_7, J_7, T, \{J_\tau\}, \{\lambda_n\}, d, \mathcal{D}_{\text{cap}}, \mathcal{H}_0^{\text{src}}, \{L, L^*\}, \\ & \mathcal{D}_{\text{fin}}, \{\mathcal{F}_s\}, \{c_s(L), c_s(L)^*\}, \text{Ev}, \text{Ev}_1, \{\mathcal{D}_{\text{ev},s}^1\}, \\ & \mathcal{E}, \mathcal{H}_0^*, \{\mathcal{H}_\tau^{\text{comp}}, B_\tau\}, \nabla^{\text{comp}}, \mathfrak{A}_7^{\text{src}}, \mathcal{R}_7^{\text{rec}}, \mathcal{F}_7, \mathcal{C}_7). \end{aligned} \quad (98)$$

The package contains the source-weight recursion, every maximal adjoint domain, the creation-adjoint coefficient, algebraic and absolute evaluation domains, parity-majorant constants, Carleman witness, auxiliary endpoint label, and first-failure record. It does not contain a nonzero relational Hamiltonian or an unrestricted completed-evaluation map.

An equivalence between  $\mathfrak{C}_7(d)$  and  $\mathfrak{C}_7(d')$  consists of source and fibre unitaries preserving the WDW-6 record, clock class, Cauchy trace, increment functionals, source vectors, grade-zero form, adjoint graphs, history words, evaluation domains, propagator, transported forms, connection, observables, records, failures, and open enlarged-carrier gates. Literal source matching requires

$$W_0 x = x \quad (x \in \mathcal{D}_{\text{cap}}). \quad (99)$$

Flow-box coordinates and response tensor frames are skins; the norm of a fixed source increment is not.

## 35. Auxiliary form plurality without physical overclassification

Fix one increment vector  $e_{n_0} \in \mathcal{D}_{\text{cap}}$ . For  $0 < a < 1$ , set

$$d_n^{(a)} = \begin{cases} (1+a)d_{n_0}, & n = n_0, \\ d_n, & n \neq n_0, \end{cases} \quad b_0^{(a)}(x, y) = \sum_{n \geq 1} d_n^{(a)} \overline{\lambda_n(x)} \lambda_n(y). \quad (100)$$

Equivalently,  $b_0^{(a)}(x, y) = b_0(x, (I + aP_{e_{n_0}})y)$ . The finite-rank operator  $I + aP_{e_{n_0}}$  and its inverse preserve every printed source-adjoint domain because  $e_{n_0}$  lies in the common star core. Thus  $b_0 \leq b_0^{(a)} \leq 2b_0$  preserves the exact radical, cap compatibility, source occupancy, density, absolute-evaluation status, and all history convergence and Carleman conclusions after changing constants. The weighted adjoint formula is recomputed with  $d^{(a)}$ .

**Theorem 35.1** (W7.18: auxiliary companion-family cardinality). *Let  $\text{Comp}_7^{\text{diag}}$  be exactly the packages eq. (98) obtained from eq. (100). Then*

$$\left| \text{Comp}_7^{\text{diag}} / \sim_{\text{comp}} \right| = 2^{\aleph_0}. \quad (101)$$

*This is a classification of auxiliary full-Cauchy Hilbert-bundle packages. It neither multiplies the singleton source-generated physical endpoint nor classifies nonzero full-Cauchy Hamiltonians.*

*Proof.* The interval  $(0, 1)$  supplies the upper and lower parameter bounds. If  $a \neq a'$ , literal source matching fixes  $e_{n_0}$ , while

$$\|e_{n_0}\|_{b_0^{(a)}}^2 = (1+a)d_{n_0} \neq (1+a')d_{n_0} = \|e_{n_0}\|_{b_0^{(a')}}^2.$$

No allowed source unitary intertwines the two forms. W7.12 shows separately that such a form and a self-adjoint star envelope do not identify a nonzero Hamiltonian with the companion derivative.  $\square$

## 36. Main theorem

**Theorem 36.1** (WDW-7 profile-clock exhaustion, Cauchy-germ evolution, and companion-extension theorem). *On the occupied Branch-A WDW-4A scalar-clock sector and accepted WDW-5/6 renormalized joint source:*

1. *the physical clock is the certified function-valued quotient class  $\phi(\cdot) = \tau\mathbf{1}$ ; every scalar-functional substitute is obstructed; and the minimal point-authentication character is unique;*
2. *the spatial mean is only an invariant analytic normal coordinate, its mean-zero complement is retained, and physical evaluation occurs at  $\chi = 0$ ;*
3. *the value/normal-response Cauchy-germ trace is continuous, quotient native, scale compatible, and nonzero on a maximal local interval;*
4. *the canonical WDW-6 physical coordinate conditions isometrically into the no-spectator line generated by the authenticated Cauchy doublet;*
5. *the renormalized WDW constraint factors through the doublet as the state-independent companion operator  $\mathbb{A}_\tau$  before the connection or Hamiltonian is defined;*
6. *restriction of that companion equation gives the Hermitian clock connection, strongly continuous unitary propagator, and full-domain bounded self-adjoint covariant Hamiltonian  $H_{\text{rel}}^C = 0$ ;*
7. *precisely the propagated sections satisfy the relational Schrödinger equation;*
8. *the fixed physical algebra  $\mathbb{CI}$  is intertwined, while the nonconstant complete volume record supplies relational change without a false fixed-operator intertwining claim;*
9. *the source-generated realization quotient is a singleton, its transported ray is stationary, and its complete-record change predicate is true;*
10. *actual cap images and frozen sections construct an explicit compatible positive grade-zero form with exact radical for the stronger full-Cauchy extension;*
11. *registered source daggers and response daggers are actual Hilbert adjoints on their printed domains;*
12. *the explicit parity majorants give a convergent response-history evolution, while unrestricted completed evaluation remains open;*
13. *the symmetric generator  $i\hbar\mathcal{K}_0^{\text{pre}}$  satisfies the Carleman condition and has a unique self-adjoint closure containing the authenticated grade-zero vector, equivalently the underlying skew envelope is essentially skew-adjoint;*
14. *companion propagation is unitary between transported Hilbert fibres;*
15. *no nonzero full-Cauchy Hamiltonian has been identified with the companion derivative;*
16. *direct source observables, residual constraints, records, and first-failure discipline are preserved; and*
17. *the auxiliary diagonal companion quotient has cardinality  $2^{\aleph_0}$  without promotion to physical multiplicity.*

*Proof.* W7.1 proves the clock. W7.2–3 prove Cauchy restriction and descent. W7.4 reapplies the renormalized constraint and proves the companion factorization. W7.5 constructs conditioning-compatible generated evolution; W7.6 constructs the fixed physical representation and complete record. W7.7–9 construct the source form and genuine star representations. W7.10 proves history evolution, W7.11 proves Carleman closure, and W7.12 gives the exact enlarged-carrier boundary. W7.13–16 carry source-observable covariance, profile discipline, change, and records. W7.17 closes the source-generated quotient; W7.18 counts only the auxiliary companion family. No conclusion uses a WDW-8 premise, a formal-dagger substitution, a point-character category switch, or a cancelling-connection claim.  $\square$

**Endpoint.** The inherited-profile-clock, source-generated WDW-7 endpoint is closed:  $|\text{Real}_7^{\text{gen}}/\sim_7| = 1$ ,  $H_{\text{rel}}^{\text{C}} = 0$  as a covariant full-fibre operator,  $\text{RayChange}_7 = \text{FALSE}$ , and  $\text{RecordChange}_7 = \text{Change}_7 = \text{TRUE}$ . The result is derived from the WDW-4A function-valued clock, complete Cauchy germ, and companion factorization, not from the scalar mean, point-character transport, or lapse selection. Its exact label is **PROFILE-CLOCK-CAUCHY-GERM-WDW7-CLOSED**. The full-Cauchy Hilbert-bundle companion construction is a stronger auxiliary extension; its nonzero-Hamiltonian identification remains open.

## 37. Load-bearing order

WDW-4A profile clock  $\longrightarrow$  scalar-functional clock exhaustion  
 $\longrightarrow$  analytic mean/mean-zero splitting without clock standing  
 $\longrightarrow$  WDW-6 physical quotient and canonical coordinate  
 $\longrightarrow$  quotient-native Cauchy-germ trace  
 $\longrightarrow$  renormalized constraint factorization  $\longrightarrow$  generated conditioning  
 $\longrightarrow$  pushed-forward form and companion-restricted connection  $\longrightarrow$  unitary propagator  
 $\longrightarrow$  self-adjoint covariant Hamiltonian  
 $\longrightarrow$  complete-record change and singleton quotient.

The independent auxiliary chain is

full response source  $\rightarrow$  compatible form  $\rightarrow$  genuine adjoints  
 $\rightarrow$  Carleman closure  $\rightarrow$  companion Hilbert bundle,

with nonzero full-Cauchy Hamiltonian identification still open. No arrow in either chain is supplied by a formal dagger, evaluation quotient, transported norm alone, point-character relabeling, or algebraically cancelling connection.

# Part VIII: Finite audit and adversarial closure

## 38. Gate ledger

| Gate  | Burden                                                                       | Status      | Basis        |
|-------|------------------------------------------------------------------------------|-------------|--------------|
| W7-00 | Frozen predecessor and component supersession boundary                       | <b>PASS</b> | Sections 1–3 |
| W7-01 | Function-valued physical-clock exhaustion and analytic-coordinate separation | <b>PASS</b> | W7.1         |

| Gate  | Burden                                                                       | Status             | Basis                    |
|-------|------------------------------------------------------------------------------|--------------------|--------------------------|
| W7-02 | Full Cauchy-germ trace, physical $\chi = 0$ evaluation, and quotient descent | <b>PASS</b>        | W7.2–3                   |
| W7-03 | State-independent constraint factorization                                   | <b>PASS</b>        | W7.4                     |
| W7-04 | WDW-6-compatible generated conditioning and pushed form                      | <b>PASS</b>        | W7.5                     |
| W7-05 | Constraint-induced connection, Hamiltonian, and propagator                   | <b>PASS</b>        | W7.5                     |
| W7-06 | Fixed physical algebra and complete-record descent                           | <b>PASS</b>        | W7.6                     |
| W7-07 | Explicit compatible grade-zero positive form                                 | <b>PASS</b>        | W7.7                     |
| W7-08 | Actual source Hilbert adjoints and dense star core                           | <b>PASS</b>        | W7.8                     |
| W7-09 | Unquotiented history star envelope and algebraic evaluation                  | <b>PASS</b>        | W7.9                     |
| W7-10 | Explicit even/odd response-history evolution                                 | <b>PASS</b>        | W7.10                    |
| W7-11 | Unrestricted completed source evaluation                                     | <b>OPEN</b>        | Stronger extension       |
| W7-12 | Genuine Carleman closure and authenticated star occupancy                    | <b>PASS</b>        | W7.11                    |
| W7-13 | Transported full-Cauchy companion Hilbert bundle                             | <b>PASS</b>        | W7.12                    |
| W7-14 | Complete-record change and singleton generated quotient                      | <b>PASS</b>        | W7.15–17                 |
| W7-15 | Nonzero full-Cauchy Hamiltonian                                              | <b>OPEN</b>        | Enlarged carrier         |
| W7-16 | Full operator-flat profile propagator                                        | <b>OPEN</b>        | Mixed profiles/curvature |
| W7-17 | Global clock and WDW-8 recovery                                              | <b>NOT-CLAIMED</b> | Section 9                |

## 39. Adversarial checks

**Audit finding 39.1** (Attack A: the point character is relabeled as a physical state). It has not. Point evaluation appears only in the authentication layer. The conditional state is the full Cauchy doublet  $Y_\tau^{(0)} = J_\tau F_{A,\mu}^{(0)}$ , and W7.5 constructs the explicit map  $\text{Cond}_\tau[u] = I_6([u])Y_\tau^{(0)}$ .

**Audit finding 39.2** (Attack B: the conditional carrier silently identifies the WDW-6 line with new Cauchy data). W7.5 prints the compatibility map and pushes the WDW-6 form through it. The codomain line lies in the quotient-native Cauchy source and is generated by the actual value/normal-response pair. The isometry is proved; it is not an identification by notation.

**Audit finding 39.3** (Attack C: point data still lose the normal derivative). The primary conditional section is  $J_\tau F_{A,\mu}^{(0)} = (R_\tau F_{A,\mu}^{(0)}, P_\tau F_{A,\mu}^{(0)})$ . Both components are retained. Point characters authenticate the source only and carry no dynamical burden.

**Audit finding 39.4** (Attack D: every differentiable section is claimed to solve a zero-Hamiltonian equation). No. Equation eq. (31) gives  $\nabla^C(\alpha Y^{(0)}) = \dot{\alpha} Y^{(0)}$ , and eq. (32) states the exact equivalence: the relational equation holds if and only if  $\dot{\alpha} = 0$ , if and only if the section is propagated.

**Audit finding 39.5** (Attack E: the zero Hamiltonian is fixed before the WDW constraint is used). The order is the reverse. W7.4 derives  $\dot{Y} = \mathbb{A}Y$  from the accepted renormalized constraint. W7.5 then

restricts  $\partial_\tau - \mathbb{A}_\tau$  to the generated line and reads off the covariant endomorphism  $H_{\text{rel}}^C = 0$ .

**Audit finding 39.6** (Attack F:  $H_{\text{rel}}^C = 0$  is the rejected cancellation in disguise). No auxiliary  $K$  is chosen and no  $\Gamma = K - \mathbb{A}$  formula occurs. The connection is the restriction of the already derived state-independent companion equation. The zero endomorphism expresses parallel transport in the induced Hilbert bundle; it does not identify  $\mathbb{A}$  with zero or with a skew-adjoint operator on the enlarged carrier.

**Audit finding 39.7** (Attack G: a changing scalar profile is claimed to intertwine). The only intertwined quantum algebra is the fixed inherited algebra  $\mathbb{C}I$ , whose scalar  $z$  is the same at both endpoints. The volume is separately typed as a clock-indexed complete record. The paper makes no equation  $U\rho_{\tau_1}(\mathcal{V}) = \rho_{\tau_2}(\mathcal{V})U$  for unequal volume values.

**Audit finding 39.8** (Attack H: the volume is merely a classical function, not a complete relational record). W7.6 defines it at the unique intersection of the occupied constraint orbit with the WDW-4A spacetime level-set gauge  $\phi(\cdot) = \tau\mathbf{1}$ . The accepted function-valued transversality theorem removes every lapse direction, WDW-3 identifies proper refoliation presentations of the same complete correlation, and proper spatial diffeomorphisms preserve total volume. The spatial mean does not appear in this proof. The result therefore factors through the declared clock and proper-redescription quotients.

**Audit finding 39.9** (Attack I: the singleton quotient is circular). The realization class is defined by antecedent source, conditioning, no-spectator, pushforward-form, companion-restriction, complete-record, and first-failure requirements. It does not include a prescribed propagator, connection coefficient, Hamiltonian value, record value, or change verdict. W7.17 derives those coordinates and then proves equivalence.

**Audit finding 39.10** (Attack J: source-generated uniqueness is promoted to uniqueness over every enrichment). It is not. W7.17 is explicitly restricted to the no-spectator line generated by the accepted WDW-6 quotient. Enriched fibres, POVMs, polarizations, additional states, and full-Cauchy Hamiltonians define different comparison classes.

**Audit finding 39.11** (Attack K: the full-Cauchy grade-zero Hilbert form is imported). W7.7 defines each cap as the actual source image, constructs a finite reduced-row-echelon section, forms the direct-limit coefficient module, and then defines the separating increments, weights, cap forms, exact radical, and completion. WDW-6 supplies serialization and finite anticipation only.

**Audit finding 39.12** (Attack L: the auxiliary form is another sampled configuration metric). The  $\lambda_n$  are frozen coefficient increments in the response source, not configuration points. Compatibility is capwise and algebraic, and no superspace measure is introduced.

**Audit finding 39.13** (Attack M: the weights are fitted only to the authenticated orbit). The family  $(z_j)$  contains every finite-cap increment vector and is saturated under every finite forward AST word before authenticated occupancy is added. The carrier is not reduced to one orbit.

**Audit finding 39.14** (Attack N: a formal transpose is called a Hilbert adjoint). Equation eq. (52) gives the maximal Hilbert-adjoint domain from the constructed form. The weighted core dagger is its restriction, and W7.8 proves the pairing and involution identities.

**Audit finding 39.15** (Attack O: row or column finiteness lacks a source reason). Column finiteness is cap locality; row finiteness is finite anticipation. The coefficient basis belongs to the constructed direct limit and is not assumed to consist of global WDW sources.

**Audit finding 39.16** (Attack P: the factorial evaluation quotient still destroys adjointness). There is no history quotient. Distinct words remain orthogonal. Creation's Hilbert adjoint is the explicitly weighted prefix-removal operator eq. (58).

**Audit finding 39.17** (Attack Q: evaluation relations are called reducing). No reducing claim is made and  $\text{Ker Ev}$  is not quotiented. W7.9 proves algebraic intertwining and the single-valued absolutely convergent extension only.

**Audit finding 39.18** (Attack R: finite anticipation is promoted to unrestricted evaluation). It is not. The absolute domain is defined by source-norm summability, and W7.10 evaluates only invariant sectors satisfying it. Unrestricted completed evaluation and authenticated absolute-domain occupancy remain open auxiliary gates.

**Audit finding 39.19** (Attack S: an equal-radius Picard estimate is used without support). W7.10 counts companion matrix paths and proves  $q(p) \leq n + 1$ . Both parity majorants and their ratio limits are printed, and the evolution loses positive analytic reserve explicitly.

**Audit finding 39.20** (Attack T: Carleman is applied before symmetry). W7.9 proves the genuine creation–annihilation adjunction first. W7.11 then defines the skew operator using the actual Hilbert adjoint before invoking the factorial estimate and Nussbaum theorem.

**Audit finding 39.21** (Attack U: the authenticated datum is called finite history). It need not be finite cap. W7.11 uses finite-cap approximants and graph convergence of the vectors and their one-power images.

**Audit finding 39.22** (Attack V: the Carleman-closed star envelope is silently identified with the WDW derivative). W7.12 expressly denies this inference. The open statement concerns the enlarged Cauchy carrier; W7.5 concerns only the constraint-generated line on which the actual evaluated Cauchy curve is already known.

**Audit finding 39.23** (Attack W: the full-Cauchy cancelling connection is merely renamed). The formula  $\Gamma = K - \mathcal{A}$  is printed and rejected. The history companion connection makes formal history solutions parallel but is not promoted to a nonzero Hamiltonian. Its unrestricted evaluated physical realization remains open.

**Audit finding 39.24** (Attack X: the continuum auxiliary family is called physical multiplicity). W7.18 counts only  $\text{Comp}_7^{\text{diag}}$ . W7.17 separately proves one class only at the source-generated no-spectator scope. No label **MANY-PHYSICAL** is used.

**Audit finding 39.25** (Attack Y: the spatial mean is treated as the physical clock). It is not. W7.1 proves that every real-valued clock functional has a nonzero kernel on the infinite-dimensional lapse distribution, while the WDW-4A profile response  $N \mapsto \epsilon_* N$  is an isomorphism. The physical gauge is  $\phi(\cdot) = \tau \mathbf{1}$ . The mean functional  $T$  appears only afterward as an invariant analytic normal coordinate and is explicitly barred from the clock and record proofs.

**Audit finding 39.26** (Attack Z: removing mean-clock standing deletes the transverse scalar response). The analytic decomposition is  $\phi = T\mathbf{1} + \chi$ , and  $\mathcal{W}_\tau$  is a germ source near  $\chi = 0$ . W7.2 retains the complete  $\chi$  germ before the physical evaluation  $\text{Ev}_\tau^\phi|_{\chi=0}$ . Tangential mean-zero functional derivatives therefore remain in  $\Theta_\tau$  and in the serialized residual grammar. Only their physical-clock status is denied.

## 40. Publication verification

The publication check confirms theorem order, function-valued physical-clock use, the scalar-functional obstruction, retention of the mean-zero scalar germ, Cauchy-conditioning compatibility, no-spectator generation, constraint-before-connection order, propagated-section equivalence, fixed-algebra intertwining, complete-record descent, and separation between source-generated physical uniqueness and auxiliary companion plurality. These finite checks verify internal coherence; they do not replace the printed proofs.

# Part IX: Exact boundary

## 41. Not claimed

The theorem does not claim:

1. a global scalar clock, preferred foliation, or global profile-time atlas;
2. that the spatial mean of  $\phi$ , any other real-valued field-space functional, or a fixed lapse is the physical refoliation clock;
3. a WDW-8 WKB, Born–Oppenheimer, Einstein, or low-energy recovery theorem;
4. that point authentication is itself the conditional state carrier;
5. that the source-generated line reconstructs arbitrary second-order WDW solution germs;
6. nontrivial projective motion of the generated physical ray;
7. a nonzero covariant Hamiltonian on the source-generated line;
8. that every differentiable line-bundle section satisfies the relational Schrödinger equation;
9. that the nonconstant volume record is one fixed quantum operator intertwined between unequal clock slices;
10. that the singleton source-generated quotient exhausts richer conditional carriers with added standing-bearing data;
11. a measure on full infinite-dimensional superspace;
12. a monolithic Hilbert completion of the entire smooth successor graph source;
13. that every direct-limit increment basis vector is itself an authenticated global WDW source;
14. a one-component positive-frequency sector or square root of  $\Theta_T$ ;
15. that the coefficient-weighted full-Cauchy grade-zero form is uniquely selected by the WDW source;
16. that evaluated source relations form a reducing subspace of the history star representation;
17. that the prefix-removal response adjoint evaluates to a pre-existing formal source dagger on every response;
18. that the Carleman-closed star envelope equals the companion derivative;
19. a nonzero source-forced Hamiltonian on the enlarged full-Cauchy carrier;
20. that continuum many compatible companion forms are continuum many physical realizations;
21. unrestricted completed evaluation or authenticated membership in the absolute evaluation domain;
22. that finite anticipation supplies a uniform analytic or Gevrey bound for the authenticated response-word series;
23. that zero-skew minimality selects a frame-independent full-Cauchy connection without a canonical source trivialization;
24. a full state-independent Tomonaga–Schwinger propagator; or
25. a companion propagator for arbitrary data outside the source-majorized evaluation domain.

## 42. Why no positive-frequency premise is needed

The primary endpoint retains the complete value/normal-response doublet of the accepted physical state and derives its first-order companion equation directly. It therefore requires no square root of a residual operator. The stronger second-order response problem likewise has an exact first-order companion

form on doubled Cauchy data and retains both frequency sectors. Positivity is required to construct the declared Hilbert carriers, not to select an invariant half-dimensional graph. No positive-frequency projection is imported.

### 43. Where the theorem can fail

The source-generated theorem fails at loss of the WDW-4A function-valued clock transversality, promotion of a scalar analytic coordinate to clock authority, deletion of the mean-zero response germ, proper quotient descent, nonzero authenticated doublet, conditioning compatibility with the WDW-6 form, companion restriction, complete-record descent, or record continuity. Those failures end that lineage. The auxiliary extension has additional failure loci: vanishing  $a_\tau$ , loss of increment separation, row or column finiteness, positive form, adjoint domain, evaluation scope, parity convergence, genuine symmetry, Carleman divergence, or companion-bundle compatibility. Mixed-profile curvature blocks the stronger profile endpoint. Unrestricted completed evaluation and the absence of a nonzero full-Cauchy Hamiltonian are extension boundaries, not defects in the narrower source-generated closure.

### 44. WDW-8 handoff

WDW-8 may begin from the unique source-generated package eq. (96) and, where useful, from each lawful auxiliary companion package eq. (98). It may investigate semiclassical recovery, enriched conditional carriers, covariantly fixed full-response connections, polarizations, or independent energy forms. Such results would extend the present endpoint. They may not rewrite the WDW-6 conditioning map, convert complete-record change into hidden ray change, or cite Carleman closure alone as identification of the full-Cauchy companion derivative with a nonzero Hamiltonian.

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